```
------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------
      name:  <unnamed>
       log:  /Users/samxie/Research/ReviewSimi_Sales/Code/outputs/core_simi_260501/logs/run_core_simi_story_exploration_260524.log
  log type:  text
 opened on:  26 May 2026, 14:46:21

. 
. di as text "Data source: `data_main'"
Data source: /Users/samxie/Research/ReviewSimi_Sales/Code/outputs/core_simi_260501/data/core_simi_panel_260501_with_mr_text_260524.dta

. di as text "Hotel profile source: `profile_csv'"
Hotel profile source: /Users/samxie/Research/ReviewSimi_Sales/Code/full-data/hotel_profile_TP.csv

. 
. *******************************************************
. ************ 1.0 Hotel Profile merge ******************
. *******************************************************
. 
. * TripAdvisor hotel profile adds product characteristics.
. * It is merged in memory only; the source panel .dta is not overwritten.
. tempfile hotel_profile

. preserve

.     import delimited using "`profile_csv'", varnames(1) clear bindquote(strict) encoding("UTF-8")
(40 vars, 11,441 obs)

.     local profile_raw_n = _N

. 
.     keep hotel_id_ta hotel_class hotel_price_low hotel_price_high hotel_room review_count ///
>         hotel_avg_rating hotel_location_rating hotel_rooms_rating hotel_value_rating ///
>         hotel_cleanliness_rating hotel_service_rating hotel_sleep_quality_rating ///
>         hotel_rank hotel_amenities hotel_style travelers_choice

. 
.     rename hotel_id_ta HotelID

.     capture confirm string variable HotelID

.     if _rc {
.         tostring HotelID, replace format(%18.0f)
HotelID was long now str8
.     }

.     replace HotelID = strtrim(HotelID)
(0 real changes made)

. 
.     duplicates tag HotelID, gen(profile_dup)

Duplicates in terms of HotelID

.     quietly count if profile_dup > 0

.     local profile_dup_n = r(N)
```

```
.     if `profile_dup_n' > 0 {
.         duplicates drop HotelID, force
.     }

.     drop profile_dup

.
.     foreach v of varlist hotel_class hotel_price_low hotel_price_high hotel_room review_count ///
>         hotel_avg_rating hotel_location_rating hotel_rooms_rating hotel_value_rating ///
>         hotel_cleanliness_rating hotel_service_rating hotel_sleep_quality_rating travelers_choice {
  2.         capture destring `v', replace ignore(",") force
  3.     }

.
.     rename hotel_class hotel_class_profile_raw

.
.     capture drop tp_price_mid ln_tp_price_mid ln_tp_room ln_tp_review_count

.     gen double tp_price_mid = (hotel_price_low + hotel_price_high) / 2 if !missing(hotel_price_low, hotel_price_high)
(4,097 missing values generated)

.     gen double ln_tp_price_mid = ln(tp_price_mid) if tp_price_mid > 0
(4,097 missing values generated)

.     gen double ln_tp_room = ln(hotel_room + 1) if !missing(hotel_room)
(975 missing values generated)

.     gen double ln_tp_review_count = ln(review_count + 1) if !missing(review_count)
(7 missing values generated)

.
.     capture drop tp_quality_index tp_service_quality

.     egen double tp_quality_index = rowmean(hotel_avg_rating hotel_location_rating hotel_rooms_rating hotel_value_rating hotel_cleanliness_rating hotel_service_rating hotel_sleep_quali
(7 missing values generated)

.     egen double tp_service_quality = rowmean(hotel_cleanliness_rating hotel_service_rating hotel_rooms_rating)
(379 missing values generated)

.
.     capture drop tp_rank_num tp_rank_total tp_rank_pct

.     gen str244 hotel_rank_short = substr(hotel_rank, 1, 244)
(1 missing value generated)

.     replace hotel_rank_short = subinstr(hotel_rank_short, ",", "", .)
(0 real changes made)

.     gen double tp_rank_num = .
(11,441 missing values generated)

.     gen double tp_rank_total = .
(11,441 missing values generated)

.     replace tp_rank_num = real(regexs(1)) if regexm(hotel_rank_short, "#([0-9]+) of ([0-9]+)")
(11,440 real changes made)

.     replace tp_rank_total = real(regexs(2)) if regexm(hotel_rank_short, "#([0-9]+) of ([0-9]+)")
(11,440 real changes made)
```

```
.     gen double tp_rank_pct = tp_rank_num / tp_rank_total if tp_rank_num > 0 & tp_rank_total > 0
(1 missing value generated)

. 
.     capture drop hotel_amenities_lc hotel_style_lc tp_amenity_count

.     gen strL hotel_amenities_lc = ustrlower(hotel_amenities)
(350 missing values generated)

.     gen strL hotel_style_lc = ustrlower(hotel_style)
(7 missing values generated)

.     gen double tp_amenity_count = 0 if !missing(hotel_amenities)
(350 missing values generated)

.     replace tp_amenity_count = strlen(hotel_amenities) - strlen(subinstr(hotel_amenities, ",", "", .)) + 1 if !missing(hotel_amenities) & strtrim(hotel_amenities) != ""
(11,091 real changes made)

. 
.     capture drop amenity_pool amenity_breakfast amenity_fitness amenity_pet amenity_business amenity_meeting

.     gen byte amenity_pool = regexm(hotel_amenities_lc, "pool") if !missing(hotel_amenities_lc)
(350 missing values generated)

.     gen byte amenity_breakfast = regexm(hotel_amenities_lc, "breakfast") if !missing(hotel_amenities_lc)
(350 missing values generated)

.     gen byte amenity_fitness = regexm(hotel_amenities_lc, "fitness|gym|workout") if !missing(hotel_amenities_lc)
(350 missing values generated)

.     gen byte amenity_pet = regexm(hotel_amenities_lc, "pet") if !missing(hotel_amenities_lc)
(350 missing values generated)

.     gen byte amenity_business = regexm(hotel_amenities_lc, "business center|business") if !missing(hotel_amenities_lc)
(350 missing values generated)

.     gen byte amenity_meeting = regexm(hotel_amenities_lc, "meeting") if !missing(hotel_amenities_lc)
(350 missing values generated)

. 
.     capture drop style_business style_family style_budget style_luxury style_modern

.     gen byte style_business = regexm(hotel_style_lc, "business") if !missing(hotel_style_lc)
(7 missing values generated)

.     gen byte style_family = regexm(hotel_style_lc, "family") if !missing(hotel_style_lc)
(7 missing values generated)

.     gen byte style_budget = regexm(hotel_style_lc, "budget|value") if !missing(hotel_style_lc)
(7 missing values generated)

.     gen byte style_luxury = regexm(hotel_style_lc, "luxury|romantic|boutique") if !missing(hotel_style_lc)
(7 missing values generated)

.     gen byte style_modern = regexm(hotel_style_lc, "modern|trendy") if !missing(hotel_style_lc)
(7 missing values generated)

. 
.     foreach v of varlist amenity_pool amenity_breakfast amenity_fitness amenity_pet amenity_business amenity_meeting ///
>         style_business style_family style_budget style_luxury style_modern {
  2.         replace `v' = 0 if missing(`v')
  3.     }
```

```
(350 real changes made)
(350 real changes made)
(350 real changes made)
(350 real changes made)
(350 real changes made)
(350 real changes made)
(7 real changes made)
(7 real changes made)
(7 real changes made)
(7 real changes made)
(7 real changes made)

.
.     capture drop travelers_choice_flag

.     gen byte travelers_choice_flag = (!missing(travelers_choice) & travelers_choice != 0)

.
.     keep HotelID hotel_class_profile_raw hotel_price_low hotel_price_high tp_price_mid ln_tp_price_mid ///
>         hotel_room ln_tp_room review_count ln_tp_review_count hotel_avg_rating hotel_location_rating ///
>         hotel_rooms_rating hotel_value_rating hotel_cleanliness_rating hotel_service_rating ///
>         hotel_sleep_quality_rating tp_quality_index tp_service_quality tp_rank_num tp_rank_total ///
>         tp_rank_pct tp_amenity_count amenity_pool amenity_breakfast amenity_fitness amenity_pet ///
>         amenity_business amenity_meeting style_business style_family style_budget style_luxury ///
>         style_modern travelers_choice_flag

.
.     save `hotel_profile', replace
(file /var/folders/x9/p1d98gj518326zdc2s5gzn8w0000gn/T//St21626.000001 not found)
file /var/folders/x9/p1d98gj518326zdc2s5gzn8w0000gn/T//St21626.000001 saved as .dta format

. restore

.
. quietly count

. local panel_raw_n = r(N)

. preserve

.     keep HotelID

.     duplicates drop

Duplicates in terms of all variables

(39,190 observations deleted)

.     local panel_hotel_n = _N

. restore

.
. merge m:1 HotelID using `hotel_profile', keep(master match) gen(profile_merge)

    Result                      Number of obs
    -----------------------------------------
    Not matched                             0
    Matched                            40,197  (profile_merge==3)
    -----------------------------------------

. quietly count if profile_merge == 3
```

```
. local profile_match_rows = r(N)

. capture drop __profile_first_hotel

. bysort HotelID: gen byte __profile_first_hotel = (_n == 1)

. quietly count if __profile_first_hotel == 1 & profile_merge == 3

. local profile_match_hotels = r(N)

. drop __profile_first_hotel

.
. capture confirm variable star_class_raw

. if _rc {
.     gen double star_class_raw = .
.     capture confirm numeric variable star_class
.     if !_rc {
.         replace star_class_raw = star_class
.     }
. }

.
. quietly count if missing(star_class_raw)

. local star_missing_before = r(N)

. quietly count if missing(star_class_raw) & !missing(hotel_class_profile_raw)

. local star_fillable = r(N)

.
. capture drop star_class_final_raw star_class_final

. gen double star_class_final_raw = star_class_raw
(21,401 missing values generated)

. replace star_class_final_raw = hotel_class_profile_raw if missing(star_class_final_raw) & !missing(hotel_class_profile_raw)
(20,067 real changes made)

. egen star_class_final = group(star_class_final_raw), label
(1,334 missing values generated)

. replace star_class_final = . if missing(star_class_final_raw)
(0 real changes made)

. label variable star_class_final_raw "Original panel star class filled by TA profile hotel_class"

. label variable star_class_final "Star class factor filled by TA profile"

.
. quietly count if missing(star_class_final_raw)

. local star_missing_after = r(N)

.
. di as text "Hotel profile audit:"
Hotel profile audit:

. di as text "  profile rows imported: `profile_raw_n'"
```

```
  profile rows imported: 11441

. di as text "  duplicate profile HotelID rows before dropping: `profile_dup_n'"
  duplicate profile HotelID rows before dropping: 0

. di as text "  panel rows matched to profile: `profile_match_rows' / `panel_raw_n'"
  panel rows matched to profile: 40197 / 40197

. di as text "  panel hotels matched to profile: `profile_match_hotels' / `panel_hotel_n'"
  panel hotels matched to profile: 1007 / 1007

. di as text "  original star_class_raw missing rows: `star_missing_before'"
  original star_class_raw missing rows: 21401

. di as text "  star_class rows fillable from profile: `star_fillable'"
  star_class rows fillable from profile: 20067

. di as text "  final star_class_final_raw missing rows: `star_missing_after'"
  final star_class_final_raw missing rows: 1334

. 
. preserve

.     clear

.     set obs 1
Number of observations (_N) was 0, now 1.

.     gen long profile_rows = `profile_raw_n'

.     gen long profile_duplicate_id_rows = `profile_dup_n'

.     gen long panel_rows = `panel_raw_n'

.     gen long panel_rows_matched = `profile_match_rows'

.     gen long panel_hotels = `panel_hotel_n'

.     gen long panel_hotels_matched = `profile_match_hotels'

.     gen long star_missing_before = `star_missing_before'

.     gen long star_fillable_from_profile = `star_fillable'

.     gen long star_missing_after = `star_missing_after'

.     gen double row_match_rate = panel_rows_matched / panel_rows

.     gen double hotel_match_rate = panel_hotels_matched / panel_hotels

.     export delimited using "`csv_dir'/story_profile_merge_audit_260524.csv", replace
file /Users/samxie/Research/ReviewSimi_Sales/Code/outputs/core_simi_260501/csv/story_profile_merge_audit_260524.csv saved

. restore

. 
. drop profile_merge

. 
. * Create numeric hotel id for FE, panel setting, and clustering.
. capture drop hotel_id_num
```

```
. capture confirm numeric variable HotelID

. if _rc {
.     encode HotelID, gen(hotel_id_num)
. }

. else {
.     gen long hotel_id_num = HotelID
. }

. 
. * Create monthly time variable.
. capture drop ym

. gen ym = monthly(year_month, "YM")

. format ym %tm

. 
. capture confirm variable Year

. if _rc {
.     gen Year = year(dofm(ym))
. }

. 
. capture confirm variable Mon

. if _rc {
.     gen Mon = month(dofm(ym))
. }

. 
. xtset hotel_id_num ym

Panel variable: hotel_id_num (unbalanced)
 Time variable: ym, 2011m1 to 2022m9, but with gaps
         Delta: 1 month

. sort hotel_id_num ym

. 
. capture drop covid2020 covid2020_2022 post2020 pre_covid

. gen byte covid2020 = (Year == 2020)

. gen byte covid2020_2022 = inrange(Year, 2020, 2022)

. gen byte post2020 = (Year >= 2020)

. gen byte pre_covid = (Year <= 2019)

. 
. ********************************************************
. ************ 1.1 winsorized revenue outcomes **********
. ********************************************************
. 
. capture drop ln_RevPAR_clean_w199 ln_lag_RevPAR_clean_w199

. gen double ln_RevPAR_clean_w199 = ln_RevPAR_clean
(714 missing values generated)
```

```
. gen double ln_lag_RevPAR_clean_w199 = ln_lag_RevPAR_clean
(1,680 missing values generated)

.
. quietly _pctile ln_RevPAR_clean if cs_sample_focus100 == 1, p(1 99)

. local y_p1 = r(r1)

. local y_p99 = r(r2)

. quietly _pctile ln_lag_RevPAR_clean if cs_sample_focus100 == 1, p(1 99)

. local ly_p1 = r(r1)

. local ly_p99 = r(r2)

.
. replace ln_RevPAR_clean_w199 = `y_p1' if ln_RevPAR_clean_w199 < `y_p1' & !missing(ln_RevPAR_clean_w199)
(430 real changes made)

. replace ln_RevPAR_clean_w199 = `y_p99' if ln_RevPAR_clean_w199 > `y_p99' & !missing(ln_RevPAR_clean_w199)
(410 real changes made)

. replace ln_lag_RevPAR_clean_w199 = `ly_p1' if ln_lag_RevPAR_clean_w199 < `ly_p1' & !missing(ln_lag_RevPAR_clean_w199)
(419 real changes made)

. replace ln_lag_RevPAR_clean_w199 = `ly_p99' if ln_lag_RevPAR_clean_w199 > `ly_p99' & !missing(ln_lag_RevPAR_clean_w199)
(397 real changes made)

.
. capture drop ln_RevPAR_clean_w195 ln_lag_RevPAR_clean_w195

. gen double ln_RevPAR_clean_w195 = ln_RevPAR_clean
(714 missing values generated)

. gen double ln_lag_RevPAR_clean_w195 = ln_lag_RevPAR_clean
(1,680 missing values generated)

.
. quietly _pctile ln_RevPAR_clean if cs_sample_focus100 == 1, p(1 95)

. local y_p1 = r(r1)

. local y_p95 = r(r2)

. quietly _pctile ln_lag_RevPAR_clean if cs_sample_focus100 == 1, p(1 95)

. local ly_p1 = r(r1)

. local ly_p95 = r(r2)

.
. replace ln_RevPAR_clean_w195 = `y_p1' if ln_RevPAR_clean_w195 < `y_p1' & !missing(ln_RevPAR_clean_w195)
(430 real changes made)

. replace ln_RevPAR_clean_w195 = `y_p95' if ln_RevPAR_clean_w195 > `y_p95' & !missing(ln_RevPAR_clean_w195)
(1,909 real changes made)

. replace ln_lag_RevPAR_clean_w195 = `ly_p1' if ln_lag_RevPAR_clean_w195 < `ly_p1' & !missing(ln_lag_RevPAR_clean_w195)
(419 real changes made)

. replace ln_lag_RevPAR_clean_w195 = `ly_p95' if ln_lag_RevPAR_clean_w195 > `ly_p95' & !missing(ln_lag_RevPAR_clean_w195)
```

```
(1,860 real changes made)

.
. capture drop ln_RevPAR_clean_w025975 ln_lag_RevPAR_clean_w025975

. gen double ln_RevPAR_clean_w025975 = ln_RevPAR_clean
(714 missing values generated)

. gen double ln_lag_RevPAR_clean_w025975 = ln_lag_RevPAR_clean
(1,680 missing values generated)

.
. quietly _pctile ln_RevPAR_clean if cs_sample_focus50 == 1, p(2.5 97.5)

. local y_p025 = r(r1)

. local y_p975 = r(r2)

. quietly _pctile ln_lag_RevPAR_clean if cs_sample_focus50 == 1, p(2.5 97.5)

. local ly_p025 = r(r1)

. local ly_p975 = r(r2)

.
. replace ln_RevPAR_clean_w025975 = `y_p025' if ln_RevPAR_clean_w025975 < `y_p025' & !missing(ln_RevPAR_clean_w025975)
(1,125 real changes made)

. replace ln_RevPAR_clean_w025975 = `y_p975' if ln_RevPAR_clean_w025975 > `y_p975' & !missing(ln_RevPAR_clean_w025975)
(1,033 real changes made)

. replace ln_lag_RevPAR_clean_w025975 = `ly_p025' if ln_lag_RevPAR_clean_w025975 < `ly_p025' & !missing(ln_lag_RevPAR_clean_w025975)
(1,082 real changes made)

. replace ln_lag_RevPAR_clean_w025975 = `ly_p975' if ln_lag_RevPAR_clean_w025975 > `ly_p975' & !missing(ln_lag_RevPAR_clean_w025975)
(1,001 real changes made)

.
. ********************************************************
. ************ 1.2 story variables and centering ********
. ********************************************************
.
. * Market-relative review flow: hotel review flow minus city-month average.
. capture drop mean_recent_cym mean_lagvol_cym rel_ln_recent_volumn rel_ln_lag_volumn_acc

. bysort CityID ym: egen mean_recent_cym = mean(ln_recent_volumn)

. bysort CityID ym: egen mean_lagvol_cym = mean(ln_lag_volumn_acc)
(87 missing values generated)

. gen double rel_ln_recent_volumn = ln_recent_volumn - mean_recent_cym

. gen double rel_ln_lag_volumn_acc = ln_lag_volumn_acc - mean_lagvol_cym
(1,007 missing values generated)

.
. * Volume nonlinearity: extra recent reviews above 10 and accumulated volume above log threshold 5.8.
. capture drop recent_above10 ln_recent_above10 recent_growth lagvol_over58 low_lagvol_58 high_lagvol_58 ln_recent_volumn_sq

. gen double recent_above10 = max(recent_volumn - 10, 0) if !missing(recent_volumn)

. gen double ln_recent_above10 = ln(recent_above10 + 1)
```

```
. gen double recent_growth = ln((recent_volumn + 1) / (lag_recent_volumn + 1)) if !missing(recent_volumn, lag_recent_volumn)
(1,007 missing values generated)

. gen double lagvol_over58 = max(ln_lag_volumn_acc - 5.8, 0) if !missing(ln_lag_volumn_acc)
(1,007 missing values generated)

. gen byte low_lagvol_58 = (ln_lag_volumn_acc < 5.8) if !missing(ln_lag_volumn_acc)
(1,007 missing values generated)

. gen byte high_lagvol_58 = (ln_lag_volumn_acc >= 5.8) if !missing(ln_lag_volumn_acc)
(1,007 missing values generated)

. gen double ln_recent_volumn_sq = ln_recent_volumn^2 if !missing(ln_recent_volumn)

.
. * Missing management-response values are zero because no lagged response activity is observed.
. foreach v of varlist lag_mr_any lag_mr_count lag_mr_rate lag_mr_text_chars lag_mr_text_words lag_mr_avg_resp_days lag_mr_med_resp_days lag_mr_quick7_share lag_mr_quick30_share lag_mr_
> share lag_mr_contact_share lag_mr_personal_share lag_mr_positive_share lag_mr_negtone_share lag_mr_template_share lag_mr_mgr_share lag_mr_neg_review_share lag_mr_neg_response_rate lag
  2.     replace `v' = 0 if missing(`v')
  3. }
(0 real changes made)
(0 real changes made)
(0 real changes made)
(0 real changes made)
(0 real changes made)
(0 real changes made)
(0 real changes made)
(0 real changes made)
(0 real changes made)
(0 real changes made)
(0 real changes made)
(0 real changes made)
(0 real changes made)
(0 real changes made)
(0 real changes made)
(0 real changes made)
(0 real changes made)
(0 real changes made)
(0 real changes made)
(0 real changes made)
(0 real changes made)
(0 real changes made)
(0 real changes made)

.
. capture drop ln_lag_mr_words ln_lag_mr_avg_words ln_lag_mr_chars

. gen double ln_lag_mr_words = ln(lag_mr_text_words + 1)

. gen double ln_lag_mr_avg_words = ln(lag_mr_avg_text_words + 1)

. gen double ln_lag_mr_chars = ln(lag_mr_text_chars + 1)

.
. * Centering keeps the original unit but makes main effects interpretable at the sample mean.
. foreach v of varlist sim_mean sim_mean_10 sim_mean_20 sim_mean_std_hotel ars_jsd_sim recent_sd rating_recent_gap rating_momentum ln_lag_avg_com_RevPAR price_gap ln_recent_volumn ln_re
> ol_over58 ln_words_acc ln_recent_volumn_sq ln_lag_mr_words ln_lag_mr_avg_words ln_lag_mr_chars lag_mr_rate lag_mr_count lag_mr_quick7_share lag_mr_apology_share lag_mr_invite_share la
> positive_share lag_mr_negtone_share lag_mr_template_share lag_mr_thanks_share lag_mr_mgr_share hotel_class_profile_raw star_class_final_raw ln_tp_price_mid ln_tp_room ln_tp_review_cou
  2.     capture drop `v'_centered
  3.     quietly summarize `v' if cs_sample_focus100 == 1 & !missing(`v')
  4.     gen double `v'_centered = `v' - r(mean) if !missing(`v')
```

```
  5. }
(223 missing values generated)
(223 missing values generated)
(223 missing values generated)
(275 missing values generated)
(223 missing values generated)
(223 missing values generated)
(1,007 missing values generated)
(1,007 missing values generated)
(1,007 missing values generated)
(1,680 missing values generated)
(1,007 missing values generated)
(1,007 missing values generated)
(1,007 missing values generated)
(1,393 missing values generated)
(1,334 missing values generated)
(2,382 missing values generated)
(148 missing values generated)
(242 missing values generated)
(9 missing values generated)

.
. * Store raw-scale distribution facts used later to translate coefficients into economic effects.
. tempname sumhandle

. tempfile sumdata

. postfile `sumhandle' str40 variable double mean sd p25 p75 using `sumdata', replace
(file /var/folders/x9/p1d98gj518326zdc2s5gzn8w0000gn/T//St21626.000005 not found)

. foreach v of varlist sim_mean sim_mean_10 sim_mean_20 sim_mean_std_hotel ars_jsd_sim ln_recent_volumn ln_lag_volumn_acc ln_words_acc lagvol_over58 lag_mr_rate lag_mr_count ln_lag_mr_w
> r_recovery_share lag_mr_positive_share lag_mr_template_share hotel_class_profile_raw star_class_final_raw ln_tp_price_mid ln_tp_room ln_tp_review_count tp_quality_index tp_service_qua
  2.     quietly summarize `v' if cs_sample_focus100 == 1, detail
  3.     post `sumhandle' ("`v'") (r(mean)) (r(sd)) (r(p25)) (r(p75))
  4. }

. postclose `sumhandle'
```

run_core_simi_story_exploration_260524.log

Generated from text log. Source: /Users/samxie/Research/ReviewSimi_Sales/Code/outputs/core_simi_260501/logs/run_core_simi_story_exploration_260524.log

```
. export delimited using "`csv_dir'/story_variable_summary_260524.csv", replace
file /Users/samxie/Research/ReviewSimi_Sales/Code/outputs/core_simi_260501/csv/story_variable_summary_260524.csv saved

. restore

.
. ********************************************************
. ************ 2. Route A: ARS as main effect ***********
. ********************************************************
.
. estimates clear

.
. * A1. Baseline ARS.
. * Core coefficient: sim_mean. Economic meaning: a 0.01 increase in ARS changes RevPAR by exp(beta*0.01)-1.
. reghdfe ln_RevPAR_clean_w199 sim_mean ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg_com_RevPAR ln_lag_RevPAR_cle
> r hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
```

```
HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  10,    557) =     128.30
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3035
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                     (Std. err. adjusted for 558 clusters in hotel_id_num)
------------------------------------------------------------------------------------------
                         |               Robust
     ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------------------+----------------------------------------------------------------
                sim_mean |  -.1846454    .057213    -3.23   0.001    -.2970249   -.0722658
       ln_recent_volumn |   .0816693   .0088687     9.21   0.000     .0642491    .0990895
              recent_sd |  -.0017433   .0075645    -0.23   0.818    -.0166017    .0131151
          recent_rating |   .0183587   .0062712     2.93   0.004     .0060406    .0306768
       ln_lag_volumn_acc |   .0291549   .0071293     4.09   0.000     .0151513    .0431586
      lag_avg_rating_acc |   -.000708   .0278115    -0.03   0.980    -.0553363    .0539204
              lag_sd_acc |  -.0167298   .0344692    -0.49   0.628    -.0844353    .0509757
    lag_avg_rating_month |   .0040355   .0017113     2.36   0.019     .0006741    .0073968
       ln_avg_com_RevPAR |   .1208847   .0120041    10.07   0.000      .097306    .1444635
ln_lag_RevPAR_clean_w199 |   .4396073   .0175846    25.00   0.000      .405067    .4741476
                   _cons |   1.481206   .1499532     9.88   0.000     1.186664    1.775749
------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store ta_sim_w199

.
. * A2. Same ARS coefficient with 1/95 upper-tail winsorization.
. reghdfe ln_RevPAR_clean_w195 sim_mean ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg_com_RevPAR ln_lag_RevPAR_cle
> r hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  10,    557) =     150.99
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8667
                                                  Adj R-squared   =     0.8638
                                                  Within R-sq.    =     0.3101
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2213

                                     (Std. err. adjusted for 558 clusters in hotel_id_num)
------------------------------------------------------------------------------------------
                         |               Robust
     ln_RevPAR_clean_w195 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------------------+----------------------------------------------------------------
                sim_mean |  -.1912741   .0545527    -3.51   0.000    -.2984283   -.0841199
       ln_recent_volumn |   .0826258   .0086273     9.58   0.000     .0656798    .0995718
              recent_sd |  -.0006678    .007335    -0.09   0.927    -.0150755    .0137399
          recent_rating |   .0182963   .0061547     2.97   0.003      .006207    .0303856
       ln_lag_volumn_acc |   .0266764   .0067893     3.93   0.000     .0133407    .0400121
```

even better

```
      lag_avg_rating_acc |  -.0034444   .0275062    -0.13   0.900    -.0574729     .050584
             lag_sd_acc |  -.0226603    .033423    -0.68   0.498    -.0883108    .0429902
    lag_avg_rating_month |   .0041216   .0016926     2.44   0.015     .0007969    .0074463
       ln_avg_com_RevPAR |   .1130516   .0109819    10.29   0.000     .0914806    .1346227
ln_lag_RevPAR_clean_w195 |   .4467287   .0163724    27.29   0.000     .4145695    .4788878
                   _cons |    1.50104   .1391614    10.79   0.000     1.227695    1.774386
------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
  Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
           ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store ta_sim_w195

.
. * A3. Same ARS coefficient with 2.5/97.5 winsorization in the tighter focus sample.
. reghdfe ln_RevPAR_clean_w025975 sim_mean ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg_com_RevPAR ln_lag_RevPAR_
> luster hotel_id_num)
(dropped 6 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     36,145
Absorbing 2 HDFE groups                           F(  10,    658) =     175.72
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8783
                                                  Adj R-squared   =     0.8755
                                                  Within R-sq.    =     0.3043
Number of clusters (hotel_id_num) =        659    Root MSE        =     0.2160

                                        (Std. err. adjusted for 659 clusters in hotel_id_num)
---------------------------------------------------------------------------------------------
                            |               Robust
    ln_RevPAR_clean_w025975 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
----------------------------+----------------------------------------------------------------
                   sim_mean |  -.1400395     .05158    -2.71   0.007    -.2413208   -.0387583
           ln_recent_volumn |   .0856702   .0080486    10.64   0.000     .0698661    .1014742
                  recent_sd |   1.43e-06   .0069406     0.00   1.000    -.0136269    .0136298
              recent_rating |   .0165202   .0056384     2.93   0.004     .0054488    .0275916
          ln_lag_volumn_acc |   .0290671   .0063386     4.59   0.000     .0166207    .0415134
         lag_avg_rating_acc |   .0018144   .0199945     0.09   0.928    -.0374463    .0410751
                 lag_sd_acc |  -.0198594   .0255512    -0.78   0.437     -.070031    .0303123
       lag_avg_rating_month |   .0037643   .0014567     2.58   0.010      .000904    .0066246
          ln_avg_com_RevPAR |   .1164576   .0109431    10.64   0.000     .0949701    .1379451
ln_lag_RevPAR_clean_w025975 |   .4345732   .0146259    29.71   0.000     .4058541    .4632924
                      _cons |   1.488761    .113447    13.12   0.000     1.265999    1.711522
---------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
  Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       659         659           0    *|
           ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store ta_sim_w025
```

```
.
. * A4. Within-hotel ARS version: tests whether months with higher within-hotel similarity have weaker revenue.
. reghdfe ln_RevPAR_clean_w199 sim_mean_std_hotel ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg_com_RevPAR ln_lag_
> vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  10,    557) =     128.03
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3035
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                     (Std. err. adjusted for 558 clusters in hotel_id_num)
------------------------------------------------------------------------------------------
                         |               Robust
    ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------------------+----------------------------------------------------------------
      sim_mean_std_hotel |  -.0065762   .0019153    -3.43   0.001    -.0103383   -.0028142
        ln_recent_volumn |    .081426   .0088657     9.18   0.000     .0640116    .0988403
               recent_sd |  -.0014399   .0074878    -0.19   0.848    -.0161478    .0132679
           recent_rating |   .0182569   .0062804     2.91   0.004     .0059208     .030593
       ln_lag_volumn_acc |   .0291026   .0071205     4.09   0.000     .0151162     .043089
      lag_avg_rating_acc |  -.0011603   .0278203    -0.04   0.967    -.0558057    .0534852
              lag_sd_acc |  -.0169374   .0345119    -0.49   0.624    -.0847267    .0508519
    lag_avg_rating_month |   .0040325   .0017121     2.36   0.019     .0006696    .0073955
       ln_avg_com_RevPAR |   .1209444   .0120113    10.07   0.000     .0973514    .1445375
ln_lag_RevPAR_clean_w199 |    .439606   .0175818    25.00   0.000     .4050713    .4741406
                   _cons |   1.429789   .1499613     9.53   0.000      1.13523    1.724347
------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store ta_hotelstd

.
. * A5. Scope-10 ARS robustness.
. reghdfe ln_RevPAR_clean_w199 sim_mean_10 ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg_com_RevPAR ln_lag_RevPAR_
> ster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  10,    557) =     128.30
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3035
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                     (Std. err. adjusted for 558 clusters in hotel_id_num)
------------------------------------------------------------------------------------------
```

no need

```
                         |               Robust
    ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------------------+----------------------------------------------------------------
             sim_mean_10 |  -.1846454    .057213    -3.23   0.001    -.2970249   -.0722658
        ln_recent_volumn |   .0816693   .0088687     9.21   0.000     .0642491    .0990895
               recent_sd |  -.0017433   .0075645    -0.23   0.818    -.0166017    .0131151
           recent_rating |   .0183587   .0062712     2.93   0.004     .0060406    .0306768
       ln_lag_volumn_acc |   .0291549   .0071293     4.09   0.000     .0151513    .0431586
      lag_avg_rating_acc |   -.000708   .0278115    -0.03   0.980    -.0553363    .0539204
              lag_sd_acc |  -.0167298   .0344692    -0.49   0.628    -.0844353    .0509757
    lag_avg_rating_month |   .0040355   .0017113     2.36   0.019     .0006741    .0073968
       ln_avg_com_RevPAR |   .1208847   .0120041    10.07   0.000      .097306    .1444635
ln_lag_RevPAR_clean_w199 |   .4396073   .0175846    25.00   0.000      .405067    .4741476
                   _cons |   1.481206   .1499532     9.88   0.000     1.186664    1.775749
------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store ta_scope10

.
. * A6. Scope-20 ARS robustness.
. reghdfe ln_RevPAR_clean_w199 sim_mean_20 ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg_com_RevPAR ln_lag_RevPAR_
> ster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  10,    557) =     127.22
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8743
                                                  Within R-sq.    =     0.3037
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2299

                                     (Std. err. adjusted for 558 clusters in hotel_id_num)
------------------------------------------------------------------------------------------
                         |               Robust
    ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------------------+----------------------------------------------------------------
             sim_mean_20 |  -.2723438   .0795619    -3.42   0.001    -.4286218   -.1160658
        ln_recent_volumn |   .0831502    .008919     9.32   0.000     .0656312    .1006693
               recent_sd |  -.0010469   .0073945    -0.14   0.887    -.0155715    .0134776
           recent_rating |   .0188691    .006225     3.03   0.003     .0066417    .0310965
       ln_lag_volumn_acc |   .0299134   .0071815     4.17   0.000     .0158073    .0440194
      lag_avg_rating_acc |   .0009861   .0278735     0.04   0.972     -.053764    .0557362
              lag_sd_acc |  -.0200448   .0345475    -0.58   0.562    -.0879041    .0478146
    lag_avg_rating_month |   .0039655   .0017113     2.32   0.021     .0006041    .0073268
       ln_avg_com_RevPAR |   .1208491   .0119933    10.08   0.000     .0972915    .1444067
ln_lag_RevPAR_clean_w199 |    .439276    .017585    24.98   0.000     .4047349     .473817
                   _cons |    1.49453   .1501549     9.95   0.000     1.199591    1.789469
------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
```

```
  Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
           ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store ta_scope20

.
. * A7. JSD-based ARS robustness.
. reghdfe ln_RevPAR_clean_w199 ars_jsd_sim ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg_com_RevPAR ln_lag_RevPAR_
> ster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  10,    557) =     129.69
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8743
                                                  Within R-sq.    =     0.3037
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2299

                                     (Std. err. adjusted for 558 clusters in hotel_id_num)
------------------------------------------------------------------------------------------
                         |               Robust
    ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------------------+----------------------------------------------------------------
             ars_jsd_sim |  -.0963536     .02523    -3.82   0.000    -.1459112   -.0467961
        ln_recent_volumn |   .0801729   .0088522     9.06   0.000     .0627851    .0975607
               recent_sd |   .0027514   .0074283     0.37   0.711    -.0118395    .0173423
           recent_rating |   .0216572     .00639     3.39   0.001     .0091059    .0342086
       ln_lag_volumn_acc |   .0299286   .0071449     4.19   0.000     .0158944    .0439628
      lag_avg_rating_acc |  -.0014094   .0277928    -0.05   0.960    -.0560009    .0531821
              lag_sd_acc |  -.0183575   .0344739    -0.53   0.595    -.0860721    .0493572
    lag_avg_rating_month |   .0040764   .0017104     2.38   0.017     .0007168    .0074361
       ln_avg_com_RevPAR |   .1210989   .0119994    10.09   0.000     .0975294    .1446685
ln_lag_RevPAR_clean_w199 |   .4393867   .0175997    24.97   0.000     .4048168    .4739567
                   _cons |   1.323043    .153577     8.61   0.000     1.021383    1.624704
------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
  Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
           ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store ta_jsd

.
. * A8. COVID boundary condition. Core interaction: ARS slope during 2020-2022 versus other months.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##i.covid2020_2022 ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg
> b(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: 1bn.covid2020_2022 is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)

HDFE Linear regression                            Number of obs   =     33,546
```

```
Absorbing 2 HDFE groups                           F(  11,    557) =     117.81
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8770
                                                  Adj R-squared   =     0.8743
                                                  Within R-sq.    =     0.3041
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2299

                                               (Std. err. adjusted for 558 clusters in hotel_id_num)
----------------------------------------------------------------------------------------------------
                                   |               Robust
                 ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-----------------------------------+----------------------------------------------------------------
                  sim_mean_centered |  -.2958544   .0635393    -4.66   0.000    -.4206603   -.1710485
                   1.covid2020_2022 |          0  (omitted)
                                   |                               降低 uncertainty
covid2020_2022#c.sim_mean_centered |
                                 1  |   .3757392   .1418047     2.65   0.008     .0972019    .6542765
                                   |
                   ln_recent_volumn |   .0814439   .0088363     9.22   0.000     .0640873    .0988005
                          recent_sd |   .0000789   .0075214     0.01   0.992    -.014695     .0148527
                      recent_rating |   .0181286   .0061963     2.93   0.004     .0059577    .0302996
                  ln_lag_volumn_acc |   .0291874   .0071355     4.09   0.000     .0151717    .0432031
                 lag_avg_rating_acc |  -.0039216   .0278017    -0.14   0.888    -.0585307    .0506875
                         lag_sd_acc |  -.0189001   .0345733    -0.55   0.585     -.08681     .0490098
               lag_avg_rating_month |   .0038451   .0017135     2.24   0.025     .0004793    .0072108
                   ln_avg_com_RevPAR |   .1202276   .0119558    10.06   0.000     .0967436    .1437115
               ln_lag_RevPAR_clean_w199 |   .4382145   .0175777    24.93   0.000     .4036879    .4727411
                              _cons |   1.451053   .1498351     9.68   0.000     1.156742    1.745364
----------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store ta_covid

.
. esttab ta_sim_w199 ta_sim_w195 ta_sim_w025 ta_hotelstd ta_scope10 ta_scope20 ta_jsd ta_covid using "`table_dir'/story_table_a_ars_main_260524.rtf", replace star(* 0.10 ** 0.05 *** 0.0
> s("Observations" "Adjusted R-squared")) mtitles("w199" "w195" "w025/975" "hotel std" "scope10" "scope20" "JSD" "COVID") nogap compress
(output written to /Users/samxie/Research/ReviewSimi_Sales/Code/outputs/core_simi_260501/tables_explicit/story_table_a_ars_main_260524.rtf)

. esttab ta_sim_w199 ta_sim_w195 ta_sim_w025 ta_hotelstd ta_scope10 ta_scope20 ta_jsd ta_covid using "`csv_dir'/story_table_a_ars_main_260524.csv", replace csv star(* 0.10 ** 0.05 *** 0
> els("Observations" "Adjusted R-squared")) mtitles("w199" "w195" "w025/975" "hotel std" "scope10" "scope20" "JSD" "COVID") nogap
(output written to /Users/samxie/Research/ReviewSimi_Sales/Code/outputs/core_simi_260501/csv/story_table_a_ars_main_260524.csv)

.
. ********************************************************
. ************ 2.1 Route A: boundary conditions *********
. ********************************************************
.
. estimates clear

.
. * A9. Rating uncertainty: if recent rating dispersion is high, ARS may matter differently.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.recent_sd_centered ln_recent_volumn recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg_com_F
> l_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
```

```
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  11,    557) =     117.73
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3035
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                                     (Std. err. adjusted for 558 clusters in hotel_id_num)
----------------------------------------------------------------------------------------------------------
                                          |               Robust
                        ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
----------------------------------------------------------------------------------------------------------
                         sim_mean_centered |  -.1788651     .05926    -3.02   0.003    -.2952656   -.0624647
                       recent_sd_centered |  -.0026415   .0077864    -0.34   0.735    -.0179359    .0126528
                                          |
c.sim_mean_centered#c.recent_sd_centered |   .0620205   .1013361     0.61   0.541    -.1370271     .261068
                                          |
                         ln_recent_volumn |   .0817325   .0088804     9.20   0.000     .0642894    .0991756
                            recent_rating |   .0180612   .0063508     2.84   0.005     .0055868    .0305356
                        ln_lag_volumn_acc |   .0290659   .0071293     4.08   0.000     .0150622    .0430695
                        lag_avg_rating_acc |  -.0003454   .0277618    -0.01   0.990    -.0548759    .0541852
                               lag_sd_acc |  -.0160793   .0343901    -0.47   0.640    -.0836294    .0514708
                      lag_avg_rating_month |   .0040196   .0017116     2.35   0.019     .0006577    .0073816
                         ln_avg_com_RevPAR |   .1208679   .0120023    10.07   0.000     .0972927    .1444431
                      ln_lag_RevPAR_clean_w199 |   .4396099   .0175881    24.99   0.000     .4050628     .474157
                                    _cons |   1.424987   .1508472     9.45   0.000     1.128688    1.721286
----------------------------------------------------------------------------------------------------------
```

无效 Inter

```
Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store am_sd

.
. * A10. Rating gap: ARS may be less useful when recent rating deviates from accumulated reputation.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.rating_recent_gap_centered ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_
>  == 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: lag_avg_rating_month omitted because of collinearity
```

路线上是：

volume $\xrightarrow{+}$ revenue

ARS $\uparrow$ −

```
HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  11,    557) =     116.87
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3035
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                                     (Std. err. adjusted for 558 clusters in hotel_id_num)
----------------------------------------------------------------------------------------------------------
                                          |               Robust
                        ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
```

```
------------------------------------------------+----------------------------------------------------------------
                              sim_mean_centered |  -.1813995     .05772    -3.14   0.002    -.2947749   -.0680241
                      rating_recent_gap_centered |   .0034715   .0018969     1.83   0.068    -.0002545    .0071975
                                                |
```

依旧不行

```
c.sim_mean_centered#c.rating_recent_gap_centered |  -.0324605   .0442271    -0.73   0.463    -.1193328    .0544117
                                                |
                               ln_recent_volumn |   .0817115   .0088819     9.20   0.000     .0642654    .0991577
                                      recent_sd |  -.0017951   .0075573    -0.24   0.812    -.0166394    .0130492
                                  recent_rating |   .0188742   .0062346     3.03   0.003      .006628    .0311205
                              ln_lag_volumn_acc |   .0291505   .0071311     4.09   0.000     .0151433    .0431578
                             lag_avg_rating_acc |   .0032095   .0277105     0.12   0.908    -.0512204    .0576394
                                     lag_sd_acc |  -.0165197   .0344706    -0.48   0.632    -.0842279    .0511885
                           lag_avg_rating_month |          0  (omitted)
                              ln_avg_com_RevPAR |   .1209029   .0120014    10.07   0.000     .0973294    .1444763
                          ln_lag_RevPAR_clean_w199 |   .4395718   .0175903    24.99   0.000     .4050204    .4741232
                                          _cons |   1.425211    .149929     9.51   0.000     1.130716    1.719707
-----------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store am_ratinggap

.
. * A11. Rating momentum: ARS may signal stability differently when recent ratings are moving.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.rating_momentum_centered ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_mc
> = 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: lag_avg_rating_month omitted because of collinearity

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  11,    557) =     116.87
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3035
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                              (Std. err. adjusted for 558 clusters in hotel_id_num)
----------------------------------------------------------------------------------------------------------------
                                               |               Robust
                          ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-----------------------------------------------+----------------------------------------------------------------
                             sim_mean_centered |  -.1813995     .05772    -3.14   0.002    -.2947749   -.0680241
                      rating_momentum_centered |   .0034715   .0018969     1.83   0.068    -.0002545    .0071975
                                               |
c.sim_mean_centered#c.rating_momentum_centered |  -.0324605   .0442271    -0.73   0.463    -.1193328    .0544117
                                               |
                              ln_recent_volumn |   .0817115   .0088819     9.20   0.000     .0642654    .0991577
                                     recent_sd |  -.0017951   .0075573    -0.24   0.812    -.0166394    .0130492
                                 recent_rating |   .0188742   .0062346     3.03   0.003      .006628    .0311205
                             ln_lag_volumn_acc |   .0291505   .0071311     4.09   0.000     .0151433    .0431578
                            lag_avg_rating_acc |   .0032095   .0277105     0.12   0.908    -.0512204    .0576394
                                    lag_sd_acc |  -.0165197   .0344706    -0.48   0.632    -.0842279    .0511885
                          lag_avg_rating_month |          0  (omitted)
```

```
                     ln_avg_com_RevPAR |   .1209029   .0120014    10.07   0.000     .0973294    .1444763
              ln_lag_RevPAR_clean_w199 |   .4395718   .0175903    24.99   0.000     .4050204    .4741232
                                 _cons |   1.425211    .149929     9.51   0.000     1.130716    1.719707
------------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store am_ratingmom

.
. * A12. Market pressure: competitor RevPAR captures local demand and competition.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.ln_lag_avg_com_RevPAR_centered ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rat
> s100 == 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
```

高 comp $\xrightarrow{+}$ revenue

$\downarrow \Delta$ ↖ -

ARS

不一定是 comp,

可能是 周边消费水平

```
HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  12,    557) =     110.67
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8811
                                                  Adj R-squared   =     0.8786
                                                  Within R-sq.    =     0.3277
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2259

                                          (Std. err. adjusted for 558 clusters in hotel_id_num)
------------------------------------------------------------------------------------------------------------------------
                                                   |               Robust
                              ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
---------------------------------------------------+--------------------------------------------------------------------
                                 sim_mean_centered |  -.1624674   .0539518    -3.01   0.003    -.2684412   -.0564936
                     ln_lag_avg_com_RevPAR_centered |  -.1540759     .01699    -9.07   0.000    -.1874482   -.1207035
                                                   |
```

X不行

```
c.sim_mean_centered#c.ln_lag_avg_com_RevPAR_centered |   .0040116   .0830824     0.05   0.962    -.1591815    .1672047
                                                   |
                                  ln_recent_volumn |   .0794886   .0084349     9.42   0.000     .0629205    .0960567
                                         recent_sd |  -.0020201   .0071315    -0.28   0.777     -.016028    .0119877
                                     recent_rating |   .0166013   .0058434     2.84   0.005     .0051235     .028079
                                 ln_lag_volumn_acc |   .0260774   .0068354     3.82   0.000     .0126511    .0395037
                                lag_avg_rating_acc |  -.0010114   .0262797    -0.04   0.969    -.0526309    .0506081
                                        lag_sd_acc |  -.0204838   .0321435    -0.64   0.524    -.0836211    .0426534
                              lag_avg_rating_month |   .0031572   .0017139     1.84   0.066    -.0002093    .0065238
                                 ln_avg_com_RevPAR |   .2272085   .0188507    12.05   0.000     .1901812    .2642357
                          ln_lag_RevPAR_clean_w199 |   .4805882    .018453    26.04   0.000     .4443421    .5168342
                                             _cons |   .8349143    .168769     4.95   0.000     .5034129    1.166416
------------------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation
```

```
. est store am_compete

.
. * A13. Product-positioning gap: price gap captures relative market position.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.price_gap_centered ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln
> bsorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  12,     557) =     110.65
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8806
                                                  Adj R-squared   =     0.8781
                                                  Within R-sq.    =     0.3249
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2264

                                     (Std. err. adjusted for 558 clusters in hotel_id_num)
--------------------------------------------------------------------------------------------
                                          |               Robust
                     ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
------------------------------------------+-------------------------------------------------
                        sim_mean_centered |  -.1623819   .0553327    -2.93   0.003    -.2710682   -.0536956
                       price_gap_centered |   .1377571   .0164132     8.39   0.000     .1055177    .1699965
                                          |
c.sim_mean_centered#c.price_gap_centered |  -.1256845   .0934749    -1.34   0.179    -.3092909     .057922
                                          |
                         ln_recent_volumn |   .0807927   .0084326     9.58   0.000     .0642291    .0973563
                                recent_sd |   -.001421   .0072383    -0.20   0.844    -.0156387    .0127967
                            recent_rating |   .0169972   .0059716     2.85   0.005     .0052677    .0287268
                        ln_lag_volumn_acc |   .0269813   .0068637     3.93   0.000     .0134994    .0404631
                       lag_avg_rating_acc |  -.0020089     .02674    -0.08   0.940    -.0545324    .0505146
                               lag_sd_acc |  -.0212926   .0327466    -0.65   0.516    -.0856145    .0430294
                     lag_avg_rating_month |     .00303   .0017312     1.75   0.081    -.0003704    .0064304
                        ln_avg_com_RevPAR |   .2162358   .0179086    12.07   0.000     .1810591    .2514125
                     ln_lag_RevPAR_clean_w199 |   .3299078   .0187657    17.58   0.000     .2930475    .3667681
                                    _cons |   1.519505   .1451695    10.47   0.000     1.234359    1.804652
--------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store am_pricegap

.
. * A14. Product quality group: star_class_final is filled from TripAdvisor profile when the panel star is missing.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean##i.star_class_final ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg_com_Re
> ar_class_final), absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: 3bn.star_class_final is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)
note: 4bn.star_class_final is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)
note: 5bn.star_class_final is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)
note: 6bn.star_class_final is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)
note: 7bn.star_class_final is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)
note: 8bn.star_class_final is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)
```

这个回去试试，

便宜酒店|贵的酒店

```
note: 9bn.star_class_final is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)

HDFE Linear regression                            Number of obs     =     32,840
Absorbing 2 HDFE groups                           F(  17,     546) =      87.46
Statistics robust to heteroskedasticity           Prob > F          =     0.0000
                                                  R-squared         =     0.8760
                                                  Adj R-squared     =     0.8733
                                                  Within R-sq.      =     0.3136
Number of clusters (hotel_id_num) =        547    Root MSE          =     0.2254

                                (Std. err. adjusted for 547 clusters in hotel_id_num)
--------------------------------------------------------------------------------------------
                            |               Robust
          ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
--------------------------------------------------------------------------------------------
                   sim_mean |   .5431269   .0770371     7.05   0.000     .3918016    .6944522
                            |
           star_class_final |
                         2  |          0  (omitted)
                       2.5  |          0  (omitted)
                         3  |          0  (omitted)
                       3.5  |          0  (omitted)
                         4  |          0  (omitted)
                       4.5  |          0  (omitted)
                         5  |          0  (omitted)
                            |
star_class_final#c.sim_mean |
                         2  |  -.5780033   .1832192    -3.15   0.002    -.9379042   -.2181024
                       2.5  |  -.8738255   .1304432    -6.70   0.000    -1.130057   -.6175935
                         3  |  -.7014467   .1068031    -6.57   0.000     -.911242   -.4916514
                       3.5  |  -.7765121   .1777792    -4.37   0.000    -1.125727   -.4272972
                         4  |  -.6423407   .1716864    -3.74   0.000    -.9795874    -.305094
                       4.5  |   -.796236   .7868567    -1.01   0.312    -2.341873    .7494009
                         5  |  -.4793303   .4809959    -1.00   0.319    -1.424159    .4654988
                            |
            ln_recent_volumn |   .0787899   .0090307     8.72   0.000     .0610506    .0965291
                  recent_sd |  -.0010955   .0076769    -0.14   0.887    -.0161753    .0139843
              recent_rating |   .0190771    .006291     3.03   0.003     .0067195    .0314347
          ln_lag_volumn_acc |   .0270932   .0072149     3.76   0.000     .0129209    .0412656
         lag_avg_rating_acc |   .0022387   .0274153     0.08   0.935    -.0516138    .0560911
                lag_sd_acc  |  -.0140983   .0344925    -0.41   0.683    -.0818526     .053656
       lag_avg_rating_month |   .0036761   .0017007     2.16   0.031     .0003354    .0070169
            ln_avg_com_RevPAR |   .1188217   .0118639    10.02   0.000     .0955172    .1421262
     ln_lag_RevPAR_clean_w199 |   .4479502   .0174617    25.65   0.000     .4136497    .4822506
                      _cons |   1.448729   .1488111     9.74   0.000     1.156416    1.741041
--------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
hotel_id_num |       547         547           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store am_star

.
. * A15. Chain product type.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##i.chain ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg_com_RevP
> n), absorb(hotel_id_num ym) vce(cluster hotel_id_num)
```

ARS本身作用可能为正？

评分低 $\xrightarrow{+}$ high uncertainty → 回归真实rating

映证低分！

ARS $\xrightarrow{-}$ high uncertainty

本意是好，为什么实际不行？ 负 → 当然↓

正 → 应当正向！

拆分

ARS真实的内容？

评论的sentiment！

文本分析 · 复杂

```
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: 1bn.chain is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)

HDFE Linear regression                            Number of obs     =     33,546
Absorbing 2 HDFE groups                           F(  11,     557) =     116.82
Statistics robust to heteroskedasticity           Prob > F          =     0.0000
                                                  R-squared         =     0.8769
                                                  Adj R-squared     =     0.8742
                                                  Within R-sq.      =     0.3035
Number of clusters (hotel_id_num) =        558    Root MSE          =     0.2300

                                      (Std. err. adjusted for 558 clusters in hotel_id_num)
-------------------------------------------------------------------------------------------
                          |               Robust
       ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
--------------------------+----------------------------------------------------------------
         sim_mean_centered |  -.1231895   .0994245    -1.24   0.216    -.3184822    .0721032
                   1.chain |          0  (omitted)
                          |
chain#c.sim_mean_centered |
                        1 |  -.0882341   .1134571    -0.78   0.437    -.3110902     .134622
                          |
          ln_recent_volumn |   .0815561   .0088756     9.19   0.000     .0641225    .0989898
                 recent_sd |  -.0016087   .0075623    -0.21   0.832    -.0164628    .0132454
             recent_rating |   .0183789   .0062645     2.93   0.003      .006074    .0306838
          ln_lag_volumn_acc |   .0292667   .0071165     4.11   0.000     .0152882    .0432452
         lag_avg_rating_acc |  -.0007967   .0278104    -0.03   0.977    -.0554229    .0538294
                lag_sd_acc |  -.0162847   .0344205    -0.47   0.636    -.0838945    .0513251
       lag_avg_rating_month |   .0040281   .0017108     2.35   0.019     .0006676    .0073886
          ln_avg_com_RevPAR |   .1208839   .0120094    10.07   0.000     .0972947    .1444731
   ln_lag_RevPAR_clean_w199 |   .4395608    .017582    25.00   0.000     .4050258    .4740958
                     _cons |   1.426324   .1498548     9.52   0.000     1.131974    1.720673
-------------------------------------------------------------------------------------------
```

数据量不行？→ 本身酒店差距大

↓

先 PSM，后DID设计

```
Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store am_chain

.
. * A16. COVID boundary repeated in the moderator table for direct comparison.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##i.covid2020_2022 ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg
> b(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: 1bn.covid2020_2022 is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)

HDFE Linear regression                            Number of obs     =     33,546
Absorbing 2 HDFE groups                           F(  11,     557) =     117.81
Statistics robust to heteroskedasticity           Prob > F          =     0.0000
                                                  R-squared         =     0.8770
                                                  Adj R-squared     =     0.8743
                                                  Within R-sq.      =     0.3041
Number of clusters (hotel_id_num) =        558    Root MSE          =     0.2299
```

```
                                              (Std. err. adjusted for 558 clusters in hotel_id_num)
----------------------------------------------------------------------------------------------------
                                   |               Robust
                ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-----------------------------------+----------------------------------------------------------------
                   sim_mean_centered |  -.2958544   .0635393    -4.66   0.000    -.4206603   -.1710485
                    1.covid2020_2022 |          0  (omitted)
                                   |
covid2020_2022#c.sim_mean_centered |
                                  1  |   .3757392   .1418047     2.65   0.008     .0972019    .6542765
                                   |
                    ln_recent_volumn |   .0814439   .0088363     9.22   0.000     .0640873    .0988005
                           recent_sd |   .0000789   .0075214     0.01   0.992     -.014695    .0148527
                       recent_rating |   .0181286   .0061963     2.93   0.004     .0059577    .0302996
                   ln_lag_volumn_acc |   .0291874   .0071355     4.09   0.000     .0151717    .0432031
                  lag_avg_rating_acc |  -.0039216   .0278017    -0.14   0.888    -.0585307    .0506875
                          lag_sd_acc |  -.0189001   .0345733    -0.55   0.585      -.08681    .0490098
                lag_avg_rating_month |   .0038451   .0017135     2.24   0.025     .0004793    .0072108
                   ln_avg_com_RevPAR |   .1202276   .0119558    10.06   0.000     .0967436    .1437115
            ln_lag_RevPAR_clean_w199 |   .4382145   .0175777    24.93   0.000     .4036879    .4727411
                               _cons |   1.451053   .1498351     9.68   0.000     1.156742    1.745364
----------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-------------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-------------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store am_covid

.
. esttab am_sd am_ratinggap am_ratingmom am_compete am_pricegap am_star am_chain am_covid using "`table_dir'/story_table_a_moderators_260524.rtf", replace star(* 0.10 ** 0.05 *** 0.01 *
> Observations" "Adjusted R-squared")) mtitles("rating sd" "rating gap" "rating mom" "competition" "price gap" "star" "chain" "COVID") nogap compress
(output written to /Users/samxie/Research/ReviewSimi_Sales/Code/outputs/core_simi_260501/tables_explicit/story_table_a_moderators_260524.rtf)

. esttab am_sd am_ratinggap am_ratingmom am_compete am_pricegap am_star am_chain am_covid using "`csv_dir'/story_table_a_moderators_260524.csv", replace csv star(* 0.10 ** 0.05 *** 0.01
> ("Observations" "Adjusted R-squared")) mtitles("rating sd" "rating gap" "rating mom" "competition" "price gap" "star" "chain" "COVID") nogap
(output written to /Users/samxie/Research/ReviewSimi_Sales/Code/outputs/core_simi_260501/csv/story_table_a_moderators_260524.csv)

.
. ********************************************************
. ************ 2.2 Route A: profile product features ****
. ********************************************************
.
. estimates clear

.
. * A17. Filled star class factor. Product main effects are absorbed by hotel FE; interactions test ARS heterogeneity.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##i.star_class_final ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_a
> issing(star_class_final), absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: 3bn.star_class_final is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)
note: 4bn.star_class_final is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)
note: 5bn.star_class_final is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)
note: 6bn.star_class_final is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)
note: 7bn.star_class_final is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)
note: 8bn.star_class_final is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)
```

```
note: 9bn.star_class_final is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)

HDFE Linear regression                            Number of obs   =     32,840
Absorbing 2 HDFE groups                           F(  17,    546) =      87.46
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8760
                                                  Adj R-squared   =     0.8733
                                                  Within R-sq.    =     0.3136
Number of clusters (hotel_id_num) =        547    Root MSE        =     0.2254

                                 (Std. err. adjusted for 547 clusters in hotel_id_num)
-------------------------------------------------------------------------------------------------------
                                     |               Robust
                ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------------------------------+-----------------------------------------------------------------
                    sim_mean_centered |   .5431269   .0770371     7.05   0.000     .3918016    .6944522
                                     |
                     star_class_final |
                                   2  |          0  (omitted)
                                 2.5  |          0  (omitted)
                                   3  |          0  (omitted)
                                 3.5  |          0  (omitted)
                                   4  |          0  (omitted)
                                 4.5  |          0  (omitted)
                                   5  |          0  (omitted)
                                     |
star_class_final#c.sim_mean_centered |
                                   2  |  -.5780033   .1832192    -3.15   0.002    -.9379042   -.2181024
                                 2.5  |  -.8738255   .1304432    -6.70   0.000    -1.130057   -.6175935
                                   3  |  -.7014467   .1068031    -6.57   0.000     -.911242   -.4916514
                                 3.5  |  -.7765121   .1777792    -4.37   0.000    -1.125727   -.4272972
                                   4  |  -.6423407   .1716864    -3.74   0.000    -.9795874    -.305094
                                 4.5  |   -.796236   .7868567    -1.01   0.312    -2.341873    .7494009
                                   5  |  -.4793303   .4809959    -1.00   0.319    -1.424159    .4654988
                                     |
                     ln_recent_volumn |   .0787899   .0090307     8.72   0.000     .0610506    .0965291
                            recent_sd |  -.0010955   .0076769    -0.14   0.887    -.0161753    .0139843
                        recent_rating |   .0190771    .006291     3.03   0.003     .0067195    .0314347
                    ln_lag_volumn_acc |   .0270932   .0072149     3.76   0.000     .0129209    .0412656
                   lag_avg_rating_acc |   .0022387   .0274153     0.08   0.935    -.0516138    .0560911
                           lag_sd_acc |  -.0140983   .0344925    -0.41   0.683    -.0818526     .053656
                 lag_avg_rating_month |   .0036761   .0017007     2.16   0.031     .0003354    .0070169
                    ln_avg_com_RevPAR |   .1188217   .0118639    10.02   0.000     .0955172    .1421262
              ln_lag_RevPAR_clean_w199 |   .4479502   .0174617    25.65   0.000     .4136497    .4822506
                                _cons |   1.398817   .1490998     9.38   0.000     1.105937    1.691696
-------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       547         547           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store ap_star_final

.
. * A18. Raw TripAdvisor hotel class. A one-unit change is roughly one star.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.hotel_class_profile_raw_centered ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_r
> cus100 == 1 & !missing(hotel_class_profile_raw), absorb(hotel_id_num ym) vce(cluster hotel_id_num)
```

```
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: hotel_class_profile_raw_centered is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)

HDFE Linear regression                            Number of obs   =     32,805
Absorbing 2 HDFE groups                           F(  11,    545) =     116.81
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8762
                                                  Adj R-squared   =     0.8735
                                                  Within R-sq.    =     0.3137
Number of clusters (hotel_id_num) =        546    Root MSE        =     0.2252

                                                                   (Std. err. adjusted for 546 clusters in hotel_id_num)
-----------------------------------------------------------------------------------------------------------------------
                                                  |               Robust
                               ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
--------------------------------------------------+--------------------------------------------------------------------
                                  sim_mean_centered |  -.1696145   .0572138    -2.96   0.003     -.282001    -.057228
                   hotel_class_profile_raw_centered |          0  (omitted)
                                                  |
c.sim_mean_centered#c.hotel_class_profile_raw_centered |  -.0194318    .088132    -0.22   0.826    -.1925518    .1536882
                                                  |
                                  ln_recent_volumn |   .0794162   .0090245     8.80   0.000     .0616892    .0971433
                                         recent_sd |  -.0010813   .0076793    -0.14   0.888    -.0161659    .0140033
                                     recent_rating |   .0192699    .006264     3.08   0.002     .0069653    .0315745
                                ln_lag_volumn_acc |   .0271446   .0072308     3.75   0.000      .012941    .0413482
                                lag_avg_rating_acc |   .0017485   .0275574     0.06   0.949    -.0523833    .0558803
                                        lag_sd_acc |  -.0145606   .0346072    -0.42   0.674    -.0825404    .0534192
                              lag_avg_rating_month |   .0036032   .0017078     2.11   0.035     .0002485    .0069578
                                ln_avg_com_RevPAR |    .118788   .0118557    10.02   0.000     .0954996    .1420764
                             ln_lag_RevPAR_clean_w199 |   .4482193   .0175043    25.61   0.000     .4138352    .4826034
                                             _cons |   1.398066   .1498693     9.33   0.000     1.103674    1.692459
-----------------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-------------------------------------------------------+
  Absorbed FE | Categories  - Redundant  = Num. Coefs |
--------------+-----------------------------------------|
 hotel_id_num |       546         546            0    *|
           ym |       135           1          134     |
-------------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store ap_hotelclass

.
. * A19. Overall profile quality index: average TripAdvisor total and sub-rating scores.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.tp_quality_index_centered ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_m
> == 1 & !missing(tp_quality_index), absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: tp_quality_index_centered is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  11,    557) =     118.27
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3035
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                                            (Std. err. adjusted for 558 clusters in hotel_id_num)
```

```
------------------------------------------------------------------------------------------------
                                               |               Robust
                              ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
------------------------------------------------+-----------------------------------------------------------------
                                 sim_mean_centered |  -.1857551   .0576198    -3.22   0.001    -.2989338   -.0725763
                          tp_quality_index_centered |          0  (omitted)
                                                   |
c.sim_mean_centered#c.tp_quality_index_centered |   .0494028   .1390259     0.36   0.722    -.2236764    .3224819
                                                   |
                                  ln_recent_volumn |   .0814721   .0088782     9.18   0.000     .0640332     .098911
                                         recent_sd |  -.0017517   .0075665    -0.23   0.817    -.0166139    .0131106
                                     recent_rating |   .0181873   .0062962     2.89   0.004     .0058202    .0305545
                                 ln_lag_volumn_acc |   .0291088   .0071397     4.08   0.000     .0150848    .0431328
                                lag_avg_rating_acc |  -.0005737   .0278648    -0.02   0.984    -.0553066    .0541592
                                        lag_sd_acc |  -.0164707   .0345822    -0.48   0.634    -.0843981    .0514567
                              lag_avg_rating_month |   .0040425   .0017142     2.36   0.019     .0006753    .0074096
                                 ln_avg_com_RevPAR |   .1208739   .0120031    10.07   0.000     .0972971    .1444507
                            ln_lag_RevPAR_clean_w199 |   .4396046   .0175822    25.00   0.000     .4050692      .47414
                                             _cons |   1.427581   .1497381     9.53   0.000     1.133461    1.721701
------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558            0    *|
          ym |       135           1          134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store ap_quality

.
. * A20. Service-quality index: cleanliness, service, and room ratings.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.tp_service_quality_centered ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating
> 0 == 1 & !missing(tp_service_quality), absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: tp_service_quality_centered is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)

HDFE Linear regression                            Number of obs   =     33,371
Absorbing 2 HDFE groups                           F(  11,    553) =     117.56
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8770
                                                  Adj R-squared   =     0.8743
                                                  Within R-sq.    =     0.3033
Number of clusters (hotel_id_num) =        554    Root MSE        =     0.2303

                                               (Std. err. adjusted for 554 clusters in hotel_id_num)
------------------------------------------------------------------------------------------------
                                                 |               Robust
                                ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
--------------------------------------------------+-----------------------------------------------------------------
                                   sim_mean_centered |  -.1834022   .0576684    -3.18   0.002    -.2966781   -.0701263
                          tp_service_quality_centered |          0  (omitted)
                                                     |
c.sim_mean_centered#c.tp_service_quality_centered |   .0420244   .1287012     0.33   0.744    -.2107787    .2948275
                                                     |
                                    ln_recent_volumn |   .0817353   .0088847     9.20   0.000     .0642835    .0991871
                                           recent_sd |  -.0017565   .0075829    -0.23   0.817    -.0166513    .0131383
                                       recent_rating |   .0180172   .0063232     2.85   0.005     .0055968    .0304376
                                   ln_lag_volumn_acc |   .0290349   .0071536     4.06   0.000     .0149833    .0430866
```

思路不对！

```
                      lag_avg_rating_acc |  -.0007729   .0279245    -0.03   0.978     -.055624    .0540782
                              lag_sd_acc |  -.0160939   .0345685    -0.47   0.642    -.0839956    .0518078
                    lag_avg_rating_month |   .0040654   .0017222     2.36   0.019     .0006826    .0074483
                       ln_avg_com_RevPAR |   .1201567   .0120405     9.98   0.000     .0965059    .1438074
                 ln_lag_RevPAR_clean_w199 |    .440203   .0176375    24.96   0.000     .4055584    .4748477
                                   _cons |   1.429067   .1501849     9.52   0.000     1.134064    1.724069
-----------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-------------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       554         554           0    *|
          ym |       135           1         134     |
-------------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store ap_service

.
. * A21. Price positioning. Product main effect is absorbed; the interaction asks whether ARS matters differently for higher-price hotels.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.ln_tp_price_mid_centered ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_mo
> = 1 & !missing(ln_tp_price_mid), absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 1 singleton observations)
(MWFE estimator converged in 7 iterations)
note: ln_tp_price_mid_centered is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)

HDFE Linear regression                            Number of obs   =     32,032
Absorbing 2 HDFE groups                           F(  11,    519) =     113.29
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8768
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3147
Number of clusters (hotel_id_num) =        520    Root MSE        =     0.2253

                                                       (Std. err. adjusted for 520 clusters in hotel_id_num)
--------------------------------------------------------------------------------------------------------------
                                               |               Robust
                          ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-----------------------------------------------+--------------------------------------------------------------
                             sim_mean_centered |  -.1602675   .0577537    -2.78   0.006    -.2737272   -.0468078
                      ln_tp_price_mid_centered |          0  (omitted)
                                               |
c.sim_mean_centered#c.ln_tp_price_mid_centered |   .1282273   .1224209     1.05   0.295    -.1122741    .3687288
                                               |
                              ln_recent_volumn |   .0800218   .0089858     8.91   0.000     .0623688    .0976749
                                     recent_sd |  -.0020791   .0077137    -0.27   0.788     -.017233    .0130747
                                 recent_rating |   .0144725   .0063755     2.27   0.024     .0019475    .0269975
                             ln_lag_volumn_acc |   .0302051   .0071887     4.20   0.000     .0160825    .0443277
                            lag_avg_rating_acc |  -.0014684    .028194    -0.05   0.958    -.0568568      .05392
                                    lag_sd_acc |  -.0207099   .0355867    -0.58   0.561    -.0906216    .0492017
                          lag_avg_rating_month |   .0040592    .001752     2.32   0.021     .0006173     .007501
                             ln_avg_com_RevPAR |   .1205647    .012223     9.86   0.000     .0965521    .1445773
                      ln_lag_RevPAR_clean_w199 |   .4457521   .0179278    24.86   0.000      .410532    .4809721
                                         _cons |   1.430159   .1524911     9.38   0.000     1.130583    1.729735
--------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-------------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       520         520           0    *|
```

```
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store ap_price

. 
. * A22. Hotel scale: number of rooms.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.ln_tp_room_centered ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month l
>  !missing(ln_tp_room), absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: ln_tp_room_centered is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)

HDFE Linear regression                            Number of obs   =     33,506
Absorbing 2 HDFE groups                           F(  11,    556) =     116.63
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8768
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3033
Number of clusters (hotel_id_num) =        557    Root MSE        =     0.2300

                                                      (Std. err. adjusted for 557 clusters in hotel_id_num)
-----------------------------------------------------------------------------------------------------------
                                          |               Robust
                     ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
------------------------------------------+----------------------------------------------------------------
                        sim_mean_centered |  -.1850537   .0576062    -3.21   0.001    -.2982061   -.0719013
                      ln_tp_room_centered |          0  (omitted)
                                          |
c.sim_mean_centered#c.ln_tp_room_centered |  -.0231578   .0724569    -0.32   0.749    -.1654806     .119165
                                          |
                         ln_recent_volumn |   .0821323   .0088273     9.30   0.000     .0647934    .0994712
                                recent_sd |  -.0009452   .0075697    -0.12   0.901     -.015814    .0139235
                            recent_rating |   .0184394   .0062899     2.93   0.004     .0060844    .0307943
                        ln_lag_volumn_acc |   .0293968   .0071811     4.09   0.000     .0152914    .0435021
                       lag_avg_rating_acc |  -.0017075   .0278922    -0.06   0.951    -.0564945    .0530795
                               lag_sd_acc |  -.0173346   .0347081    -0.50   0.618    -.0855098    .0508405
                     lag_avg_rating_month |   .0040677   .0017171     2.37   0.018      .000695    .0074404
                        ln_avg_com_RevPAR |   .1207604    .012003    10.06   0.000     .0971836    .1443371
                  ln_lag_RevPAR_clean_w199 |   .4394204   .0176124    24.95   0.000     .4048255    .4740154
                                    _cons |   1.429103   .1503042     9.51   0.000     1.133869    1.724336
-----------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
hotel_id_num |       557         557           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store ap_room

. 
. * A23. TripAdvisor rank percentile within destination/type list. Lower values mean better profile rank.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.tp_rank_pct_centered ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month
> & !missing(tp_rank_pct), absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: tp_rank_pct_centered is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)
```

```
HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  11,    557) =     117.79
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3036
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                                       (Std. err. adjusted for 558 clusters in hotel_id_num)
------------------------------------------------------------------------------------------------------------
                                           |               Robust
                      ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------------------------------------+----------------------------------------------------------------
                         sim_mean_centered |  -.1882112   .0574819    -3.27   0.001    -.3011189   -.0753035
                      tp_rank_pct_centered |          0  (omitted)
                                           |
c.sim_mean_centered#c.tp_rank_pct_centered |  -.2579506   .2365054    -1.09   0.276    -.7225021    .2066009
                                           |
                          ln_recent_volumn |   .0808458   .0089336     9.05   0.000     .0632981    .0983936
                                 recent_sd |  -.0018895   .0075853    -0.25   0.803    -.0167888    .0130097
                             recent_rating |   .0177013   .0062855     2.82   0.005      .005355    .0300475
                         ln_lag_volumn_acc |   .0290061   .0071442     4.06   0.000     .0149732    .0430389
                        lag_avg_rating_acc |  -.0005752   .0278908    -0.02   0.984    -.0553593    .0542088
                                lag_sd_acc |  -.0162197   .0346419    -0.47   0.640    -.0842645    .0518252
                      lag_avg_rating_month |   .0040609   .0017125     2.37   0.018     .0006971    .0074247
                         ln_avg_com_RevPAR |   .1208588   .0120023    10.07   0.000     .0972834    .1444342
                  ln_lag_RevPAR_clean_w199 |   .4395638   .0175708    25.02   0.000     .4050507    .4740768
                                     _cons |   1.431561   .1492346     9.59   0.000      1.13843    1.724692
------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-------------------------------------------------------+
  Absorbed FE | Categories  - Redundant  = Num. Coefs |
--------------+----------------------------------------|
 hotel_id_num |       558         558           0    *|
           ym |       135           1         134     |
-------------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store ap_rank

.
. * A24. Amenity breadth from profile amenities text.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.tp_amenity_count_centered ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_m
> == 1 & !missing(tp_amenity_count), absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: tp_amenity_count_centered is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  11,    557) =     118.94
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3035
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                                            (Std. err. adjusted for 558 clusters in hotel_id_num)
-----------------------------------------------------------------------------------------------------------------
                                                |               Robust
                           ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
```

```
-------------------------------------------------+----------------------------------------------------------------
                             sim_mean_centered |  -.1843556   .0575388    -3.20   0.001    -.2973752   -.0713361
                     tp_amenity_count_centered |          0  (omitted)
                                                |
c.sim_mean_centered#c.tp_amenity_count_centered |   .0005199   .0059972     0.09   0.931      -.01126    .0122998
                                                |
                              ln_recent_volumn |   .0815613    .008829     9.24   0.000     .0642191    .0989036
                                     recent_sd |  -.0017612   .0075851    -0.23   0.816    -.0166601    .0131377
                                 recent_rating |   .0183291   .0062769     2.92   0.004     .0059998    .0306584
                              ln_lag_volumn_acc |   .0291454   .0071452     4.08   0.000     .0151105    .0431803
                             lag_avg_rating_acc |  -.0006774   .0278232    -0.02   0.981    -.0553286    .0539737
                                    lag_sd_acc |  -.0166922   .0345072    -0.48   0.629    -.0844724    .0510879
                           lag_avg_rating_month |   .0040376   .0017133     2.36   0.019     .0006724    .0074029
                              ln_avg_com_RevPAR |   .1208777    .012002    10.07   0.000      .097303    .1444525
                        ln_lag_RevPAR_clean_w199 |   .4396035   .0175773    25.01   0.000     .4050777    .4741294
                                         _cons |   1.427379   .1495919     9.54   0.000     1.133546    1.721212
--------------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store ap_amenity

.
. * A25. Travelers' Choice badge. Missing badge field is coded as no badge in the profile.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##i.travelers_choice_flag ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month
>  & !missing(travelers_choice_flag), absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: 0bn.travelers_choice_flag is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)
note: 0.travelers_choice_flag omitted because of collinearity

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  10,    557) =     128.30
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3035
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                                     (Std. err. adjusted for 558 clusters in hotel_id_num)
------------------------------------------------------------------------------------------------------------
                                         |               Robust
                    ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-----------------------------------------+------------------------------------------------------------------
                       sim_mean_centered |  -.1846454    .057213    -3.23   0.001    -.2970249   -.0722658
                 0.travelers_choice_flag |          0  (omitted)
                                         |
travelers_choice_flag#c.sim_mean_centered |
                                       0 |          0  (omitted)
                                         |
                        ln_recent_volumn |   .0816693   .0088687     9.21   0.000     .0642491    .0990895
                               recent_sd |  -.0017433   .0075645    -0.23   0.818    -.0166017    .0131151
                           recent_rating |   .0183587   .0062712     2.93   0.004     .0060406    .0306768
                        ln_lag_volumn_acc |   .0291549   .0071293     4.09   0.000     .0151513    .0431586
                       lag_avg_rating_acc |   -.000708   .0278115    -0.03   0.980    -.0553363    .0539204
```

不对！
这里应重做

```
                       lag_sd_acc |  -.0167298   .0344692    -0.49   0.628    -.0844353    .0509757
             lag_avg_rating_month |   .0040355   .0017113     2.36   0.019     .0006741    .0073968
                ln_avg_com_RevPAR |   .1208847   .0120041    10.07   0.000      .097306    .1444635
          ln_lag_RevPAR_clean_w199 |   .4396073   .0175846    25.00   0.000      .405067    .4741476
                            _cons |      1.427   .1498731     9.52   0.000     1.132614    1.721386
------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-------------------------------------------------------+
  Absorbed FE | Categories  - Redundant  = Num. Coefs |
--------------+----------------------------------------|
 hotel_id_num |       558         558           0    *|
           ym |       135           1         134     |
-------------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store ap_choice

.
. esttab ap_star_final ap_hotelclass ap_quality ap_service ap_price ap_room ap_rank ap_amenity ap_choice using "`table_dir'/story_table_a_profile_product_260524.rtf", replace star(* 0.1
> ats(N r2_a, labels("Observations" "Adjusted R-squared")) mtitles("star final" "TA class" "quality" "service" "price" "rooms" "rank pct" "amenities" "choice") nogap compress
(output written to /Users/samxie/Research/ReviewSimi_Sales/Code/outputs/core_simi_260501/tables_explicit/story_table_a_profile_product_260524.rtf)

. esttab ap_star_final ap_hotelclass ap_quality ap_service ap_price ap_room ap_rank ap_amenity ap_choice using "`csv_dir'/story_table_a_profile_product_260524.csv", replace csv star(* 0
> stats(N r2_a, labels("Observations" "Adjusted R-squared")) mtitles("star final" "TA class" "quality" "service" "price" "rooms" "rank pct" "amenities" "choice") nogap
(output written to /Users/samxie/Research/ReviewSimi_Sales/Code/outputs/core_simi_260501/csv/story_table_a_profile_product_260524.csv)

.
. ********************************************************
. ************ 2.3 Route A: profile flags ***************
. ********************************************************
```

各个特质做 inter 意义不大，
其实可以根据分类聚类再来做！

增大样本

有理论依据

```
.
. estimates clear

.
. * A26. Pool amenity boundary.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##i.amenity_pool ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg_c
> ng(amenity_pool), absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: 1bn.amenity_pool is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  11,    557) =     116.54
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3035
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                           (Std. err. adjusted for 558 clusters in hotel_id_num)
-------------------------------------------------------------------------------------------------
                                  |               Robust
             ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
----------------------------------+--------------------------------------------------------------
                sim_mean_centered |  -.2212055   .1234436    -1.79   0.074    -.4636774    .0212664
                   1.amenity_pool |          0  (omitted)
                                  |
amenity_pool#c.sim_mean_centered |
                                1 |   .0428544   .1354431     0.32   0.752    -.2231872     .308896
                                  |
                 ln_recent_volumn |   .0817367   .0089005     9.18   0.000      .064254    .0992195
```

```
                 recent_sd |  -.0018288   .0075603    -0.24   0.809     -.016679    .0130213
             recent_rating |   .0183562   .0062721     2.93   0.004     .0060363    .0306762
         ln_lag_volumn_acc |   .0291559   .0071289     4.09   0.000     .0151531    .0431587
        lag_avg_rating_acc |  -.0004731   .0277664    -0.02   0.986    -.0550128    .0540667
                lag_sd_acc |  -.0167196   .0344606    -0.49   0.628    -.0844082    .0509691
      lag_avg_rating_month |   .0040338   .0017108     2.36   0.019     .0006734    .0073942
         ln_avg_com_RevPAR |   .1208936   .0120032    10.07   0.000     .0973165    .1444707
     ln_lag_RevPAR_clean_w199 |   .4396096   .0175854    25.00   0.000     .4050678    .4741515
                     _cons |   1.425994    .149745     9.52   0.000      1.13186    1.720128
---------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-------------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+-----------------------------------------|
 hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-------------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store af_pool

.
. * A27. Breakfast amenity boundary.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##i.amenity_breakfast ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_
> missing(amenity_breakfast), absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: 1bn.amenity_breakfast is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  11,    557) =     117.15
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3036
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                                  (Std. err. adjusted for 558 clusters in hotel_id_num)
--------------------------------------------------------------------------------------------------------
                                      |               Robust
                 ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
--------------------------------------+-----------------------------------------------------------------
                    sim_mean_centered |   .0511499   .1948117     0.26   0.793    -.3315055    .4338053
                  1.amenity_breakfast |          0  (omitted)
                                      |
amenity_breakfast#c.sim_mean_centered |
                                    1 |  -.2570523    .199987    -1.29   0.199    -.6498731    .1357686
                                      |
                     ln_recent_volumn |   .0813586   .0088936     9.15   0.000     .0638895    .0988278
                            recent_sd |  -.0013416   .0075807    -0.18   0.860    -.0162318    .0135486
                        recent_rating |   .0185595   .0062976     2.95   0.003     .0061897    .0309294
                    ln_lag_volumn_acc |   .0292406   .0071294     4.10   0.000     .0152368    .0432443
                   lag_avg_rating_acc |  -.0009787   .0276267    -0.04   0.972    -.0552439    .0532866
                           lag_sd_acc |  -.0167715   .0343675    -0.49   0.626    -.0842773    .0507342
                 lag_avg_rating_month |   .0040479   .0017108     2.37   0.018     .0006876    .0074083
                    ln_avg_com_RevPAR |   .1210256    .012004    10.08   0.000      .097447    .1446041
             ln_lag_RevPAR_clean_w199 |   .4395263   .0175836    25.00   0.000     .4049881    .4740645
                                _cons |   1.426663   .1491272     9.57   0.000     1.133742    1.719583
--------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
```

```
------------------------------------------------------+
  Absorbed FE | Categories  - Redundant  = Num. Coefs |
--------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
           ym |       135           1         134     |
------------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store af_breakfast

.
. * A28. Fitness amenity boundary.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##i.amenity_fitness ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_av
> ssing(amenity_fitness), absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: 1bn.amenity_fitness is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  11,    557) =     116.65
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3035
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                                (Std. err. adjusted for 558 clusters in hotel_id_num)
----------------------------------------------------------------------------------------------------
                                    |               Robust
               ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
------------------------------------+---------------------------------------------------------------
                  sim_mean_centered |  -.1812707   .1467247    -1.24   0.217    -.4694722    .1069307
                  1.amenity_fitness |          0  (omitted)
                                    |
amenity_fitness#c.sim_mean_centered |
                                  1 |  -.0040571   .1554737    -0.03   0.979    -.3094436    .3013293
                                    |
                   ln_recent_volumn |   .0816726   .0088578     9.22   0.000     .0642738    .0990713
                          recent_sd |  -.001734    .0075632    -0.23   0.819    -.0165898    .0131218
                      recent_rating |   .0183653   .0062855     2.92   0.004     .0060191    .0307114
                  ln_lag_volumn_acc |   .0291585   .0071408     4.08   0.000     .0151323    .0431848
                 lag_avg_rating_acc |  -.0007081     .027812    -0.03   0.980    -.0553372    .0539211
                         lag_sd_acc |  -.0167143   .0344369    -0.49   0.628    -.0843564    .0509278
               lag_avg_rating_month |    .004035   .0017129     2.36   0.019     .0006705    .0073996
                   ln_avg_com_RevPAR |   .1208865   .0120051    10.07   0.000     .0973057    .1444672
           ln_lag_RevPAR_clean_w199 |   .4396084   .0175845    25.00   0.000     .4050684    .4741484
                              _cons |   1.426892   .1499331     9.52   0.000     1.132388    1.721395
----------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
------------------------------------------------------+
  Absorbed FE | Categories  - Redundant  = Num. Coefs |
--------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
           ym |       135           1         134     |
------------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store af_fitness

.
. * A29. Pet-friendly amenity boundary.
```

```
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##i.amenity_pet ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg_co
> g(amenity_pet), absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: 1bn.amenity_pet is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  11,    557) =     116.59
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3036
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                (Std. err. adjusted for 558 clusters in hotel_id_num)
-------------------------------------------------------------------------------------
                              |               Robust
         ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
------------------------------+------------------------------------------------------
            sim_mean_centered |   -.295347   .0882149    -3.35   0.001    -.4686216   -.1220724
                1.amenity_pet |          0  (omitted)
                              |
amenity_pet#c.sim_mean_centered |
                            1 |   .1648279   .1063265     1.55   0.122    -.0440221    .3736779
                              |
             ln_recent_volumn |   .0813735   .0088658     9.18   0.000      .063959    .0987879
                    recent_sd |  -.0018155   .0075773    -0.24   0.811    -.0166991     .013068
                recent_rating |   .0181613   .0062758     2.89   0.004     .0058341    .0304884
            ln_lag_volumn_acc |    .029074   .0071485     4.07   0.000     .0150326    .0431154
           lag_avg_rating_acc |  -.0011726   .0279407    -0.04   0.967    -.0560546    .0537094
                   lag_sd_acc |  -.0170801   .0345657    -0.49   0.621    -.0849751     .050815
         lag_avg_rating_month |   .0040713   .0017127     2.38   0.018     .0007072    .0074355
             ln_avg_com_RevPAR |   .1208323   .0119937    10.07   0.000      .097274    .1443906
       ln_lag_RevPAR_clean_w199 |   .4395622   .0175777    25.01   0.000     .4050356    .4740889
                        _cons |   1.431796   .1505128     9.51   0.000     1.136154    1.727438
-------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store af_pet

.
. * A30. Business-center amenity boundary.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##i.amenity_business ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_a
> issing(amenity_business), absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: 1bn.amenity_business is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  11,    557) =     117.20
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3035
```

接近

```
Number of clusters (hotel_id_num) =        558     Root MSE        =     0.2300

                                                  (Std. err. adjusted for 558 clusters in hotel_id_num)
-------------------------------------------------------------------------------------------------------
                                      |               Robust
                  ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
--------------------------------------+----------------------------------------------------------------
                      sim_mean_centered |  -.1738363   .1199476    -1.45   0.148    -.4094412    .0617685
                     1.amenity_business |          0  (omitted)
                                      |
amenity_business#c.sim_mean_centered |
                                    1  |  -.0141237   .1296565    -0.11   0.913    -.2687992    .2405517
                                      |
                       ln_recent_volumn |   .0816518    .008887     9.19   0.000     .0641957    .0991078
                              recent_sd |  -.0016899   .0075462    -0.22   0.823    -.0165124    .0131327
                          recent_rating |    .018378   .0062757     2.93   0.004      .006051     .030705
                      ln_lag_volumn_acc |    .029174   .0071439     4.08   0.000     .0151417    .0432064
                     lag_avg_rating_acc |  -.0007195   .0278135    -0.03   0.979    -.0553517    .0539127
                             lag_sd_acc |  -.0167286   .0344703    -0.49   0.628    -.0844363     .050979
                   lag_avg_rating_month |   .0040355   .0017112     2.36   0.019     .0006744    .0073967
                       ln_avg_com_RevPAR |   .1208888   .0120048    10.07   0.000     .0973086    .1444691
               ln_lag_RevPAR_clean_w199 |   .4396018   .0175873    25.00   0.000     .4050563    .4741473
                                  _cons |    1.42681    .149868     9.52   0.000     1.132434    1.721185
-------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store af_business

.
. * A31. Meeting-room amenity boundary.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##i.amenity_meeting ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_av
> ssing(amenity_meeting), absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: 1bn.amenity_meeting is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  11,     557) =     117.06
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3035
Number of clusters (hotel_id_num) =        558     Root MSE        =     0.2300

                                                 (Std. err. adjusted for 558 clusters in hotel_id_num)
------------------------------------------------------------------------------------------------------
                                     |               Robust
                 ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------------------------------+----------------------------------------------------------------
                    sim_mean_centered |   -.164266   .1218156    -1.35   0.178    -.4035401    .0750081
                    1.amenity_meeting |          0  (omitted)
                                     |
amenity_meeting#c.sim_mean_centered |
                                   1  |  -.0261292   .1335471    -0.20   0.845    -.2884467    .2361883
```

这两就可以合并成一类

```
                                     |
                   ln_recent_volumn |   .0817003   .0088627     9.22   0.000     .0642919    .0991087
                          recent_sd |  -.0016494   .0075417    -0.22   0.827     -.016463    .0131641
                      recent_rating |   .0184206   .0062819     2.93   0.004     .0060814    .0307598
                   ln_lag_volumn_acc |   .0291818   .0071388     4.09   0.000     .0151595    .0432042
                  lag_avg_rating_acc |  -.0007995    .027795    -0.03   0.977    -.0553953    .0537963
                          lag_sd_acc |  -.0168337   .0344906    -0.49   0.626    -.0845813    .0509139
                lag_avg_rating_month |   .0040302   .0017149     2.35   0.019     .0006619    .0073986
                    ln_avg_com_RevPAR |   .1208933   .0120066    10.07   0.000     .0973096     .144477
                ln_lag_RevPAR_clean_w199 |   .4395987   .0175881    24.99   0.000     .4050515    .4741458
                               _cons |   1.426818   .1498447     9.52   0.000     1.132488    1.721148
-----------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-------------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
           ym |       135           1         134     |
-------------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store af_meeting

.
. * A32. Business-style profile boundary.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##i.style_business ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg
> sing(style_business), absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: 1bn.style_business is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  11,     557) =     117.13
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8743
                                                  Within R-sq.    =     0.3037
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2299

                                              (Std. err. adjusted for 558 clusters in hotel_id_num)
----------------------------------------------------------------------------------------------------
                                  |               Robust
             ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
----------------------------------+-----------------------------------------------------------------
                sim_mean_centered |  -.2023136   .0574473    -3.52   0.000    -.3151533   -.0894738
                 1.style_business |          0  (omitted)
                                  |
style_business#c.sim_mean_centered |
                                1 |   .6821942   .4808632     1.42   0.157    -.2623327    1.626721
                                  |
                 ln_recent_volumn |   .0817155   .0088474     9.24   0.000     .0643372    .0990938
                        recent_sd |  -.0013798   .0075411    -0.18   0.855    -.0161923    .0134326
                    recent_rating |   .0184387   .0062734     2.94   0.003     .0061162    .0307612
                ln_lag_volumn_acc |    .029258   .0070631     4.14   0.000     .0153845    .0431315
               lag_avg_rating_acc |  -.0015618   .0278566    -0.06   0.955    -.0562787    .0531551
                       lag_sd_acc |  -.0165299   .0345134    -0.48   0.632    -.0843222    .0512625
             lag_avg_rating_month |   .0040281   .0017104     2.36   0.019     .0006685    .0073877
                ln_avg_com_RevPAR |    .120687   .0120235    10.04   0.000       .09707    .1443041
         ln_lag_RevPAR_clean_w199 |   .4393025    .017535    25.05   0.000     .4048597    .4737454
                            _cons |   1.430539   .1501215     9.53   0.000     1.135665    1.725413
----------------------------------------------------------------------------------------------------
```

与前面合并即可

```
Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store af_style_business

.
. * A33. Family-style profile boundary.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##i.style_family ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg_c
> ng(style_family), absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: 1bn.style_family is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  11,    557) =     117.22
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3035
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                             (Std. err. adjusted for 558 clusters in hotel_id_num)
--------------------------------------------------------------------------------------------------
                                 |               Robust
            ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
---------------------------------+----------------------------------------------------------------
               sim_mean_centered |  -.1876861   .0654396    -2.87   0.004    -.3162246   -.0591476
                  1.style_family |          0  (omitted)
                                 |
style_family#c.sim_mean_centered |
                               1 |   .0139168   .1069751     0.13   0.897    -.1962071    .2240407
                                 |
                ln_recent_volumn |   .0816798   .0088651     9.21   0.000     .0642667     .099093
                       recent_sd |  -.0017597   .0075675    -0.23   0.816    -.0166241    .0131047
                   recent_rating |    .018344   .0062782     2.92   0.004     .0060121    .0306758
               ln_lag_volumn_acc |   .0291669   .0071371     4.09   0.000     .0151479    .0431859
              lag_avg_rating_acc |   -.000727   .0278028    -0.03   0.979    -.0553381    .0538841
                      lag_sd_acc |  -.0167329   .0344715    -0.49   0.628    -.0844429    .0509772
            lag_avg_rating_month |   .0040358   .0017114     2.36   0.019     .0006742    .0073974
               ln_avg_com_RevPAR |   .1208794   .0120034    10.07   0.000      .097302    .1444569
        ln_lag_RevPAR_clean_w199 |    .439603   .0175881    24.99   0.000      .405056    .4741501
                           _cons |   1.427101   .1498586     9.52   0.000     1.132744    1.721458
--------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store af_style_family
```

```
.
. * A34. Budget-style profile boundary.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##i.style_budget ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg_c
> ng(style_budget), absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: 1bn.style_budget is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  11,     557) =     117.07
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3035
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                             (Std. err. adjusted for 558 clusters in hotel_id_num)
---------------------------------------------------------------------------------------------------
                                  |               Robust
             ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
----------------------------------+----------------------------------------------------------------
                sim_mean_centered |  -.1783892   .0612785    -2.91   0.004    -.2987544   -.0580239
                   1.style_budget |          0  (omitted)
                                  |
style_budget#c.sim_mean_centered |
                                1 |  -.0311489   .1367374    -0.23   0.820     -.299733    .2374352
                                  |
                 ln_recent_volumn |   .0816497   .0088671     9.21   0.000     .0642327    .0990667
                        recent_sd |  -.0017814   .0075688    -0.24   0.814    -.0166483    .0130854
                    recent_rating |   .0183335    .006278     2.92   0.004      .006002    .0306649
                ln_lag_volumn_acc |    .029177   .0071278     4.09   0.000     .0151764    .0431775
               lag_avg_rating_acc |   -.000573   .0278028    -0.02   0.984    -.0551841    .0540382
                       lag_sd_acc |  -.0165393   .0344807    -0.48   0.632    -.0842674    .0511888
             lag_avg_rating_month |   .0040376   .0017129     2.36   0.019     .0006729    .0074022
                ln_avg_com_RevPAR |   .1208667   .0120005    10.07   0.000     .0972949    .1444384
         ln_lag_RevPAR_clean_w199 |   .4395933     .01758    25.01   0.000     .4050622    .4741244
                            _cons |   1.426518   .1499421     9.51   0.000     1.131997    1.721039
---------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store af_style_budget

.
. * A35. Luxury/romantic/boutique-style profile boundary. → 合并为一类即可
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##i.style_luxury ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg_c
> ng(style_luxury), absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: 1bn.style_luxury is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  11,     557) =     116.88
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
```

```
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3035
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                             (Std. err. adjusted for 558 clusters in hotel_id_num)
---------------------------------------------------------------------------------------------------
                                 |               Robust
              ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
---------------------------------+-----------------------------------------------------------------
                 sim_mean_centered |  -.1796825   .0579729    -3.10   0.002    -.2935547   -.0658102
                    1.style_luxury |          0  (omitted)
                                 |
style_luxury#c.sim_mean_centered |
                               1  |  -.0929795   .2810278    -0.33   0.741    -.6449834    .4590245
                                 |
                  ln_recent_volumn |   .0817857   .0088856     9.20   0.000     .0643323    .0992391
                         recent_sd |  -.0017827   .0075552    -0.24   0.814    -.0166228    .0130574
                     recent_rating |   .0184276   .0062683     2.94   0.003     .0061151      .03074
                 ln_lag_volumn_acc |   .0291223   .0071307     4.08   0.000     .0151158    .0431287
                lag_avg_rating_acc |  -.0007566   .0278132    -0.03   0.978    -.0553883     .053875
                        lag_sd_acc |  -.0168931   .0344938    -0.49   0.625    -.0846468    .0508607
              lag_avg_rating_month |    .004026   .0017118     2.35   0.019     .0006635    .0073884
                 ln_avg_com_RevPAR |   .1209105   .0120042    10.07   0.000     .0973315    .1444894
            ln_lag_RevPAR_clean_w199 |    .439609   .0175854    25.00   0.000     .4050673    .4741507
                             _cons |   1.426947   .1498803     9.52   0.000     1.132547    1.721347
---------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
  Absorbed FE | Categories  - Redundant  = Num. Coefs |
--------------+--------------------------------------|
 hotel_id_num |       558         558           0    *|
           ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store af_style_luxury

.
. * A36. Modern/trendy-style profile boundary.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##i.style_modern ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg_c
> ng(style_modern), absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
note: 1bn.style_modern is probably collinear with the fixed effects (all partialled-out values are close to zero; tol = 1.0e-09)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  11,    557) =     117.25
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3035
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                             (Std. err. adjusted for 558 clusters in hotel_id_num)
---------------------------------------------------------------------------------------------------
                                 |               Robust
              ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
---------------------------------+-----------------------------------------------------------------
                 sim_mean_centered |  -.1571667   .0674069    -2.33   0.020    -.2895695   -.0247639
                    1.style_modern |          0  (omitted)
                                 |
```

```
style_modern#c.sim_mean_centered |
                               1  |  -.1025255   .1010785    -1.01   0.311    -.3010672    .0960162
                                  |
                 ln_recent_volumn |   .0815459    .008887     9.18   0.000     .0640898     .099002
                        recent_sd |  -.0017722   .0075639    -0.23   0.815    -.0166294     .013085
                    recent_rating |   .0184067   .0062716     2.93   0.003     .0060879    .0307255
                ln_lag_volumn_acc |   .0291103   .0071224     4.09   0.000     .0151202    .0431004
               lag_avg_rating_acc |  -.0009523   .0278246    -0.03   0.973    -.0556062    .0537016
                       lag_sd_acc |  -.0171589    .034469    -0.50   0.619     -.084864    .0505462
             lag_avg_rating_month |   .0040375    .001711     2.36   0.019     .0006767    .0073983
                ln_avg_com_RevPAR |   .1207816    .012015    10.05   0.000     .0971814    .1443819
         ln_lag_RevPAR_clean_w199 |   .4395325   .0175821    25.00   0.000     .4049971    .4740679
                            _cons |   1.429632   .1500637     9.53   0.000     1.134873    1.724392
---------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
------------------------------------------------------+
  Absorbed FE | Categories  - Redundant  = Num. Coefs |
--------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
           ym |       135           1         134     |
------------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store af_style_modern

.
. esttab af_pool af_breakfast af_fitness af_pet af_business af_meeting af_style_business af_style_family af_style_budget af_style_luxury af_style_modern using "`table_dir'/story_table_a
> 0.001) cells(b(star fmt(4)) se(par fmt(4))) stats(N r2_a, labels("Observations" "Adjusted R-squared")) mtitles("pool" "breakfast" "fitness" "pet" "business amenity" "meeting" "busines
> nogap compress
(output written to /Users/samxie/Research/ReviewSimi_Sales/Code/outputs/core_simi_260501/tables_explicit/story_table_a_profile_flags_260524.rtf)

. esttab af_pool af_breakfast af_fitness af_pet af_business af_meeting af_style_business af_style_family af_style_budget af_style_luxury af_style_modern using "`csv_dir'/story_table_a_p
> * 0.001) cells(b(star fmt(4)) se(par fmt(4))) stats(N r2_a, labels("Observations" "Adjusted R-squared")) mtitles("pool" "breakfast" "fitness" "pet" "business amenity" "meeting" "busin
> ) nogap
(output written to /Users/samxie/Research/ReviewSimi_Sales/Code/outputs/core_simi_260501/csv/story_table_a_profile_flags_260524.csv)

.
. *******************************************************
. ************ 3. Route B: volume x ARS *****************
. *******************************************************
```

⇒ 这个其实就是探索前面的间接.

```
.
. estimates clear

.
. * B1. Recent review flow. Core interaction: recent monthly volume x ARS.
. * A doubling of recent volume is ln(2); ARS is interpreted in 0.01 raw-unit increments.
. reghdfe ln_RevPAR_clean c.ln_recent_volumn_centered##c.sim_mean_centered recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg_com_RevPAR
> m) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  11,     557) =      78.19
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8569
                                                  Adj R-squared   =     0.8538
                                                  Within R-sq.    =     0.2520
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2654

                                                            (Std. err. adjusted for 558 clusters in hotel_id_num)
------------------------------------------------------------------------------------------------------------------
```

```
                                          |               Robust
                             ln_RevPAR_clean | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
---------------------------------------------+----------------------------------------------------------------
                    ln_recent_volumn_centered |    .089775   .0106741     8.41   0.000     .0688087    .1107413
                            sim_mean_centered |  -.2263089   .0671774    -3.37   0.001    -.3582609   -.0943568
                                             |
c.ln_recent_volumn_centered#c.sim_mean_centered |  -.3171372   .1661671    -1.91   0.057    -.6435279    .0092535
```

调可显著

很对，越多越相似越不好！

```
                                             |
                                    recent_sd |  -.0044735   .0089491    -0.50   0.617    -.0220516    .0131046
                                recent_rating |   .0233843    .007151     3.27   0.001     .0093381    .0374304
                           ln_lag_volumn_acc |   .0350554   .0083362     4.21   0.000     .0186812    .0514296
                          lag_avg_rating_acc |  -.0035013   .0316978    -0.11   0.912    -.0657631    .0587606
                                  lag_sd_acc |  -.0195034   .0377764    -0.52   0.606    -.0937049    .0546981
                        lag_avg_rating_month |   .0031816   .0022623     1.41   0.160    -.0012622    .0076254
                           ln_avg_com_RevPAR |    .130275   .0137021     9.51   0.000     .1033609     .157189
                          ln_lag_RevPAR_clean |   .3972821   .0286658    13.86   0.000     .3409759    .4535884
                                       _cons |   1.758062   .1763871     9.97   0.000     1.411597    2.104527
--------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store tb_recent

.
```

不调

```
. * B2. Recent review pressure above 10 reviews in the month.
. reghdfe ln_RevPAR_clean_w199 c.ln_recent_above10_centered##c.sim_mean_centered recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg_com_
> el_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  11,     557) =     122.37
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8770
                                                  Adj R-squared   =     0.8744
                                                  Within R-sq.    =     0.3046
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2298

                                                   (Std. err. adjusted for 558 clusters in hotel_id_num)
---------------------------------------------------------------------------------------------------------------
                                              |               Robust
                         ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
----------------------------------------------+----------------------------------------------------------------
                   ln_recent_above10_centered |   .0421703   .0040854    10.32   0.000     .0341456     .050195
                            sim_mean_centered |  -.2002585   .0573615    -3.49   0.001    -.3129299   -.0875872
                                              |
c.ln_recent_above10_centered#c.sim_mean_centered |  -.0844799   .0644223    -1.31   0.190    -.2110203    .0420606
                                              |
                                    recent_sd |  -.0019772   .0075451    -0.26   0.793    -.0167975    .0128432
                                recent_rating |   .0178275   .0062611     2.85   0.005     .0055293    .0301257
                            ln_lag_volumn_acc |    .028696   .0071036     4.04   0.000     .0147428    .0426491
                           lag_avg_rating_acc |  -.0023312   .0277469    -0.08   0.933    -.0568326    .0521701
                                   lag_sd_acc |  -.0152403   .0344189    -0.44   0.658    -.0828471    .0523665
                         lag_avg_rating_month |   .0039887   .0017177     2.32   0.021     .0006147    .0073627
                            ln_avg_com_RevPAR |   .1200897   .0120085    10.00   0.000     .0965023    .1436772
```

```
                         ln_lag_RevPAR_clean_w199 |   .4390518   .0176355    24.90   0.000     .4044116     .473692
                                            _cons |   1.665091   .1520893    10.95   0.000     1.366352    1.963829
-----------------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-------------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+-----------------------------------------|
 hotel_id_num |       558         558           0    *|
           ym |       135           1         134     |
-------------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store tb_recent10

.
. * B3. Review growth from last month.
```

只看增长量，故事不好讲，不用！

```
. reghdfe ln_RevPAR_clean_w195 c.recent_growth_centered##c.sim_mean_centered recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg_com_RevP
> d_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  11,    557) =     115.77
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8645
                                                  Adj R-squared   =     0.8616
                                                  Within R-sq.    =     0.2991
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2231

                                                         (Std. err. adjusted for 558 clusters in hotel_id_num)
-------------------------------------------------------------------------------------------------------------
                                        |               Robust
                   ln_RevPAR_clean_w195 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
----------------------------------------+--------------------------------------------------------------------
                 recent_growth_centered |   .0799697   .0067113    11.92   0.000     .0667872    .0931523
                      sim_mean_centered |  -.1133552   .0561399    -2.02   0.044    -.2236269   -.0030834
                                        |
c.recent_growth_centered#c.sim_mean_centered |  -.2426722   .1418447    -1.71   0.088    -.5212882    .0359437
                                        |
                              recent_sd |  -.0017245   .0075845    -0.23   0.820    -.0166223    .0131733
                          recent_rating |   .0216507    .006235     3.47   0.001     .0094036    .0338977
                      ln_lag_volumn_acc |   .0436221   .0071057     6.14   0.000     .0296648    .0575794
                     lag_avg_rating_acc |   .0023668   .0281687     0.08   0.933    -.052963     .0576967
                             lag_sd_acc |  -.0192574    .034058    -0.57   0.572    -.0861551    .0476404
                   lag_avg_rating_month |   .0041657   .0017011     2.45   0.015     .0008243    .0075071
                      ln_avg_com_RevPAR |    .113629   .0108854    10.44   0.000     .0922475    .1350105
               ln_lag_RevPAR_clean_w199 |   .4318169   .0161431    26.75   0.000      .400108    .4635258
                                  _cons |   1.588364   .1414544    11.23   0.000     1.310514    1.866213
-------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-------------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+-----------------------------------------|
 hotel_id_num |       558         558           0    *|
           ym |       135           1         134     |
-------------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store tb_growth
```

```
.
. * B4. Recent review flow relative to city-month competitors.
. reghdfe ln_RevPAR_clean_w199 c.rel_ln_recent_volumn_centered##c.sim_mean_centered recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg_c
> hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  11,    557) =     117.71
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3035
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                                               (Std. err. adjusted for 558 clusters in hotel_id_num)
----------------------------------------------------------------------------------------------------------------------
                                                 |               Robust
                            ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------------------------------------------+--------------------------------------------------------------------
                      rel_ln_recent_volumn_centered |   .0827891    .008898     9.30   0.000     .0653114    .1002668
                               sim_mean_centered |  -.1855506   .0573607    -3.23   0.001    -.2982204   -.0728808
                                                 |
c.rel_ln_recent_volumn_centered#c.sim_mean_centered |  -.0998779   .1474287    -0.68   0.498     -.389462    .1897063
                                                 |
                                       recent_sd |  -.0017783   .0075612    -0.24   0.814    -.0166302    .0130736
                                   recent_rating |   .0182796   .0062587     2.92   0.004     .0059861    .0305731
                               ln_lag_volumn_acc |   .0293035   .0071414     4.10   0.000     .0152762    .0433308
                              lag_avg_rating_acc |  -.0009145   .0277318    -0.03   0.974    -.0553862    .0535573
                                      lag_sd_acc |  -.0172333   .0344591    -0.50   0.617     -.084919    .0504524
                            lag_avg_rating_month |   .0040263   .0017118     2.35   0.019     .0006639    .0073887
                               ln_avg_com_RevPAR |   .1209975   .0120206    10.07   0.000     .0973863    .1446087
                            ln_lag_RevPAR_clean_w199 |   .4396328   .0175914    24.99   0.000     .4050793    .4741863
                                           _cons |   1.649375   .1522441    10.83   0.000     1.350332    1.948417
----------------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
           ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store tb_relrecent

.
. * B5. Accumulated review stock. Treat this as an alternative volume-stock measure rather than controlling for another volume stock.
. reghdfe ln_RevPAR_clean_w199 c.ln_lag_volumn_acc##c.sim_mean_centered recent_sd recent_rating lag_avg_rating_acc lag_sd_acc ln_avg_com_RevPAR ln_lag_RevPAR_clean_w199 if cs_sample_foc
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(   9,    557) =     117.53
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8762
                                                  Adj R-squared   =     0.8736
                                                  Within R-sq.    =     0.2998
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2306

                                                   (Std. err. adjusted for 558 clusters in hotel_id_num)
```

WHY?

累积历史评论很多 → 大酒店 → revenue 高

+

缓解 ⇒ 评论数量作为背书

近期相似 ARS

−

```
---------------------------------------------------------------------------------------------------------
                                         |               Robust
                       ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
---------------------------------------------------------------------------------------------------------
                        ln_lag_volumn_acc |   .0349716   .0070779     4.94   0.000     .021069    .0488742
                        sim_mean_centered |  -.7658154   .2535958    -3.02   0.003   -1.263936   -.2676945
                                          |
c.ln_lag_volumn_acc#c.sim_mean_centered |   .1218487   .0471931     2.58   0.010    .0291505    .2145468
                                          |
                                recent_sd |  -.0004229   .0076227    -0.06   0.956   -.0153957    .0145499
                            recent_rating |   .0250736   .0060078     4.17   0.000    .0132728    .0368743
                       lag_avg_rating_acc |   .0062127   .0275059     0.23   0.821   -.0478153    .0602407
                               lag_sd_acc |  -.0166674   .0342266    -0.49   0.626   -.0838964    .0505615
                        ln_avg_com_RevPAR |   .1216856   .0120495    10.10   0.000    .0980176    .1453536
                   ln_lag_RevPAR_clean_w199 |   .4442004   .0174381    25.47   0.000    .4099478    .4784529
                                    _cons |   1.554122   .1486233    10.46   0.000    1.262191    1.846052
---------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store tb_cum

.
. * B6. Accumulated review stock with within-hotel ARS. This checks whether the volume story survives an alternative ARS measure.
. reghdfe ln_RevPAR_clean_w199 c.ln_lag_volumn_acc_centered##c.sim_mean_std_hotel_centered recent_sd recent_rating lag_avg_rating_acc lag_sd_acc ln_avg_com_RevPAR ln_lag_RevPAR_clean_w1
> el_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
                                                              Yes it did
HDFE Linear regression                           Number of obs   =     33,546
Absorbing 2 HDFE groups                          F(   9,    557) =     117.09
Statistics robust to heteroskedasticity          Prob > F        =     0.0000
                                                 R-squared       =     0.8762
                                                 Adj R-squared   =     0.8736
                                                 Within R-sq.    =     0.2997
Number of clusters (hotel_id_num) =        558   Root MSE        =     0.2306

                                                       (Std. err. adjusted for 558 clusters in hotel_id_num)
-------------------------------------------------------------------------------------------------------------------------
                                                        |               Robust
                                   ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------------------------------------------------------------------------------------------------------------------
                              ln_lag_volumn_acc_centered |   .0361026   .0070557     5.12   0.000    .0222435    .0499617
                             sim_mean_std_hotel_centered |  -.0038189   .0019284    -1.98   0.048   -.0076067   -.0000311
                                                        |
c.ln_lag_volumn_acc_centered#c.sim_mean_std_hotel_centered |   .0040439   .0015627     2.59   0.010    .0009743    .0071134
                                                        |
                                              recent_sd |  -.0011189   .0075456    -0.15   0.882   -.0159403    .0137024
                                          recent_rating |   .0253517   .0060308     4.20   0.000    .0135057    .0371976
                                     lag_avg_rating_acc |    .007956   .0276745     0.29   0.774   -.0464032    .0623151
                                             lag_sd_acc |  -.0171643   .0345009    -0.50   0.619   -.0849321    .0506035
                                      ln_avg_com_RevPAR |   .1220229   .0120778    10.10   0.000    .0982993    .1457464
                                 ln_lag_RevPAR_clean_w199 |   .4443267   .0174294    25.49   0.000    .4100914    .4785621
                                                  _cons |   1.739606   .1568603    11.09   0.000    1.431496    2.047716
-------------------------------------------------------------------------------------------------------------------------
```

```
Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store tb_cum_hstd

.
. * B7. Accumulated review stock above log threshold 5.8.   中位数
. reghdfe ln_RevPAR_clean_w199 c.lagvol_over58_centered##c.sim_mean_centered recent_sd recent_rating lag_avg_rating_acc lag_sd_acc ln_avg_com_RevPAR ln_lag_RevPAR_clean_w199 if cs_sampl
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(   9,    557) =     103.96
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8759
                                                  Adj R-squared   =     0.8733
                                                  Within R-sq.    =     0.2981
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2309

                                         (Std. err. adjusted for 558 clusters in hotel_id_num)
-----------------------------------------------------------------------------------------------
                                          |               Robust
                     ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
------------------------------------------+----------------------------------------------------
                   lagvol_over58_centered |  -.0012507   .0114168    -0.11   0.913     -.023676    .0211747
                        sim_mean_centered |  -.0607946   .0569094    -1.07   0.286    -.1725778    .0509886
                                          |
c.lagvol_over58_centered#c.sim_mean_centered |   .3075802   .1104995     2.78   0.006     .0905335     .524627
                                          |
                                recent_sd |  -.0004112   .0075985    -0.05   0.957    -.0153364    .0145141
                            recent_rating |   .0235606   .0060111     3.92   0.000     .0117534    .0353677
                       lag_avg_rating_acc |   .0264426   .0274153     0.96   0.335    -.0274073    .0802926
                               lag_sd_acc |   .0087954    .033874     0.26   0.795    -.0577411    .0753319
                        ln_avg_com_RevPAR |   .1219003   .0120861    10.09   0.000     .0981604    .1456402
                 ln_lag_RevPAR_clean_w199 |   .4502711   .0171405    26.27   0.000     .4166032    .4839391
                                    _cons |   1.619768   .1496239    10.83   0.000     1.325872    1.913664
-----------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store tb_cum58

.
. * B8. Accumulated review stock above log threshold 5.8 with scope-10 ARS.
. reghdfe ln_RevPAR_clean_w199 c.lagvol_over58_centered##c.sim_mean_10_centered recent_sd recent_rating lag_avg_rating_acc lag_sd_acc ln_avg_com_RevPAR ln_lag_RevPAR_clean_w199 if cs_sa
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
```

```
HDFE Linear regression                              Number of obs   =     33,546
Absorbing 2 HDFE groups                             F(   9,    557) =     103.96
Statistics robust to heteroskedasticity             Prob > F        =     0.0000
                                                    R-squared       =     0.8759
                                                    Adj R-squared   =     0.8733
                                                    Within R-sq.    =     0.2981
Number of clusters (hotel_id_num) =        558     Root MSE        =     0.2309

                                                              (Std. err. adjusted for 558 clusters in hotel_id_num)
-------------------------------------------------------------------------------------------------------------------
                                               |               Robust
                             ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------------------------------------------+-----------------------------------------------------------------
                          lagvol_over58_centered |  -.0012507   .0114168    -0.11   0.913     -.023676    .0211747
                            sim_mean_10_centered |  -.0607946   .0569094    -1.07   0.286    -.1725778    .0509886
                                                 |
c.lagvol_over58_centered#c.sim_mean_10_centered |   .3075802   .1104995     2.78   0.006     .0905335     .524627
                                                 |
                                       recent_sd |  -.0004112   .0075985    -0.05   0.957    -.0153364    .0145141
                                   recent_rating |   .0235606   .0060111     3.92   0.000     .0117534    .0353677
                              lag_avg_rating_acc |   .0264426   .0274153     0.96   0.335    -.0274073    .0802926
                                      lag_sd_acc |   .0087954    .033874     0.26   0.795    -.0577411    .0753319
                               ln_avg_com_RevPAR |   .1219003   .0120861    10.09   0.000     .0981604    .1456402
                            ln_lag_RevPAR_clean_w199 |   .4502711   .0171405    26.27   0.000     .4166032    .4839391
                                           _cons |   1.619768   .1496239    10.83   0.000     1.325872    1.913664
-------------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store tb_cum58_s10

.                                                                              理解有误
. * B9. Accumulated review text stock. This captures solicitation/content intensity beyond count.
. reghdfe ln_RevPAR_clean_w199 c.ln_words_acc_centered##c.sim_mean_centered recent_sd recent_rating lag_avg_rating_acc lag_sd_acc ln_avg_com_RevPAR ln_lag_RevPAR_clean_w199 if cs_sample
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                              Number of obs   =     33,546
Absorbing 2 HDFE groups                             F(   9,    557) =     115.92
Statistics robust to heteroskedasticity             Prob > F        =     0.0000
                                                    R-squared       =     0.8763
                                                    Adj R-squared   =     0.8737
                                                    Within R-sq.    =     0.3004
Number of clusters (hotel_id_num) =        558     Root MSE        =     0.2305

                                                         (Std. err. adjusted for 558 clusters in hotel_id_num)
---------------------------------------------------------------------------------------------------------------
                                           |               Robust
                      ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
---------------------------------------------+-----------------------------------------------------------------
                     ln_words_acc_centered |   .0480579   .0088135     5.45   0.000     .0307462    .0653697
                         sim_mean_centered |  -.0798092   .0571843    -1.40   0.163    -.1921324     .032514
                                             |
c.ln_words_acc_centered#c.sim_mean_centered |   .1173592   .0489518     2.40   0.017     .0212065    .2135118
                                             |
```

```
                         recent_sd |   .0003384   .0076371     0.04   0.965    -.0146626    .0153393
                     recent_rating |   .0253585    .006028     4.21   0.000     .0135181    .0371988
                lag_avg_rating_acc |   .0067711   .0274225     0.25   0.805    -.0470931    .0606354
                        lag_sd_acc |  -.0231884   .0345813    -0.67   0.503    -.0911141    .0447373
                  ln_avg_com_RevPAR |   .1216738   .0120666    10.08   0.000     .0979723    .1453754
           ln_lag_RevPAR_clean_w199 |   .4432677   .0174156    25.45   0.000     .4090595     .477476
                             _cons |   1.754158   .1551375    11.31   0.000     1.449432    2.058884
--------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-------------------------------------------------------+
  Absorbed FE | Categories  - Redundant  = Num. Coefs |
--------------+----------------------------------------|
 hotel_id_num |       558         558           0    *|
           ym |       135           1         134     |
-------------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store tb_words

.
. * B10. Text stock with within-hotel ARS.
. reghdfe ln_RevPAR_clean_w199 c.ln_words_acc_centered##c.sim_mean_std_hotel_centered recent_sd recent_rating lag_avg_rating_acc lag_sd_acc ln_avg_com_RevPAR ln_lag_RevPAR_clean_w199 if
> _num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(   9,    557) =     115.79
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8763
                                                  Adj R-squared   =     0.8737
                                                  Within R-sq.    =     0.3004
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2305

                                                          (Std. err. adjusted for 558 clusters in hotel_id_num)
-----------------------------------------------------------------------------------------------------------------------
                                                    |               Robust
                               ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
----------------------------------------------------+------------------------------------------------------------------
                              ln_words_acc_centered |   .0489937    .008758     5.59   0.000     .0317909    .0661965
                        sim_mean_std_hotel_centered |  -.0034432   .0019145    -1.80   0.073    -.0072036    .0003173
                                                    |
c.ln_words_acc_centered#c.sim_mean_std_hotel_centered |   .0042229   .0016295     2.59   0.010     .0010221    .0074237
                                                    |
                                          recent_sd |  -.0003038   .0075601    -0.04   0.968    -.0151537     .014546
                                      recent_rating |   .0255947   .0060486     4.23   0.000     .0137138    .0374756
                                 lag_avg_rating_acc |   .0086269   .0275618     0.31   0.754    -.0455109    .0627647
                                         lag_sd_acc |  -.0235909   .0347576    -0.68   0.498    -.0918628     .044681
                                  ln_avg_com_RevPAR |    .121962   .0120878    10.09   0.000     .0982188    .1457052
                           ln_lag_RevPAR_clean_w199 |   .4432986   .0174067    25.47   0.000     .4091079    .4774894
                                              _cons |   1.745704   .1556633    11.21   0.000     1.439945    2.051463
-----------------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-------------------------------------------------------+
  Absorbed FE | Categories  - Redundant  = Num. Coefs |
--------------+----------------------------------------|
 hotel_id_num |       558         558           0    *|
           ym |       135           1         134     |
-------------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation
```

```
. est store tb_words_hstd

.
. * B11. Nonlinear recent volume. Core terms are the linear and squared volume interactions with ARS.
. reghdfe ln_RevPAR_clean_w199 ln_recent_volumn_centered ln_recent_volumn_sq_centered sim_mean_centered c.ln_recent_volumn_centered#c.sim_mean_centered c.ln_recent_volumn_sq_centered#c.
> c lag_avg_rating_month ln_avg_com_RevPAR ln_lag_RevPAR_clean_w199 if cs_sample_focus100 == 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                              Number of obs   =     33,546
Absorbing 2 HDFE groups                             F(  12,    557) =     101.63
Statistics robust to heteroskedasticity             Prob > F        =     0.0000
                                                    R-squared       =     0.8768
                                                    Adj R-squared   =     0.8742
                                                    Within R-sq.    =     0.3032
Number of clusters (hotel_id_num) =        558      Root MSE        =     0.2300

                                                                 (Std. err. adjusted for 558 clusters in hotel_id_num)
--------------------------------------------------------------------------------------------------------------------
                                             |               Robust
                        ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
---------------------------------------------+----------------------------------------------------------------------
                   ln_recent_volumn_centered |   .3227955   .0491495     6.57   0.000     .2262545    .4193366
                ln_recent_volumn_sq_centered |  -.0386222   .0078218    -4.94   0.000    -.0539859   -.0232585
                           sim_mean_centered |  -.1714549   .0564624    -3.04   0.003    -.2823602   -.0605496
                                             |
   c.ln_recent_volumn_centered#c.sim_mean_centered |  -1.038811   .8617815    -1.21   0.229     -2.73155    .6539282
                                             |
c.ln_recent_volumn_sq_centered#c.sim_mean_centered |   .1545724     .13892     1.11   0.266    -.1182988    .4274436
                                             |
                                   recent_sd |  -.0014348   .0075136    -0.19   0.849    -.0161934    .0133237
                               recent_rating |   .0171014   .0062648     2.73   0.007     .0047959     .029407
                           lag_avg_rating_acc |   .0081662   .0277567     0.29   0.769    -.0463544    .0626869
                                  lag_sd_acc |  -.0081643   .0345711    -0.24   0.813      -.07607    .0597414
                         lag_avg_rating_month |   .0040008   .0017226     2.32   0.021     .0006172    .0073844
                            ln_avg_com_RevPAR |   .1202301   .0120405     9.99   0.000     .0965798    .1438804
                     ln_lag_RevPAR_clean_w199 |   .4437358   .0173854    25.52   0.000     .4095868    .4778848
                                       _cons |   1.755664   .1545171    11.36   0.000     1.452156    2.059171
--------------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store tb_recent_sq

.
. esttab tb_recent tb_recent10 tb_growth tb_relrecent tb_cum tb_cum_hstd tb_cum58 tb_cum58_s10 tb_words tb_words_hstd tb_recent_sq using "`table_dir'/story_table_b_volume_ars_260524.rtf
> )) se(par fmt(4))) stats(N r2_a, labels("Observations" "Adjusted R-squared")) mtitles("recent" "recent>10" "growth" "rel recent" "cumulative" "cum hstd" "cum>5.8" "cum>5.8 s10" "text
(output written to /Users/samxie/Research/ReviewSimi_Sales/Code/outputs/core_simi_260501/tables_explicit/story_table_b_volume_ars_260524.rtf)

. esttab tb_recent tb_recent10 tb_growth tb_relrecent tb_cum tb_cum_hstd tb_cum58 tb_cum58_s10 tb_words tb_words_hstd tb_recent_sq using "`csv_dir'/story_table_b_volume_ars_260524.csv",
> (4)) se(par fmt(4))) stats(N r2_a, labels("Observations" "Adjusted R-squared")) mtitles("recent" "recent>10" "growth" "rel recent" "cumulative" "cum hstd" "cum>5.8" "cum>5.8 s10" "tex
(output written to /Users/samxie/Research/ReviewSimi_Sales/Code/outputs/core_simi_260501/csv/story_table_b_volume_ars_260524.csv)

.
```

Not proved

This is Solicitation

```
. *********************************************************
. ************ 4. Route C: reply -> revenue *************
. *********************************************************
.
. estimates clear

.
. * C1. Any reply. Direct coefficient is the revenue gap between hotels with and without lagged reply activity at average ARS.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##i.lag_mr_any ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg_com
> tel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  12,    557) =     109.21
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3036
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                           (Std. err. adjusted for 558 clusters in hotel_id_num)
------------------------------------------------------------------------------------------------
                               |               Robust
          ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------------------------+----------------------------------------------------------------
             sim_mean_centered |   -.207918   .0691757    -3.01   0.003     -.343795   -.0720409
                  1.lag_mr_any |   .0062976   .0046346     1.36   0.175    -.0028058    .0154009
                               |
lag_mr_any#c.sim_mean_centered |
                             1 |   .0507799    .081928     0.62   0.536    -.1101458    .2117055
                               |
              ln_recent_volumn |   .0806223   .0087452     9.22   0.000     .0634448    .0977998
                     recent_sd |   -.001932   .0075698    -0.26   0.799    -.0168008    .0129368
                 recent_rating |   .0180882   .0062777     2.88   0.004     .0057573    .0304192
             ln_lag_volumn_acc |   .0290609   .0071492     4.06   0.000     .0150182    .0431036
            lag_avg_rating_acc |  -.0005777   .0278398    -0.02   0.983    -.0552616    .0541061
                    lag_sd_acc |  -.0170498   .0345171    -0.49   0.622    -.0848494    .0507497
          lag_avg_rating_month |   .0041972   .0017048     2.46   0.014     .0008485    .0075459
            ln_avg_com_RevPAR |    .120919    .012021    10.06   0.000     .0973069     .144531
      ln_lag_RevPAR_clean_w199 |   .4393458   .0176397    24.91   0.000     .4046973    .4739943
                         _cons |   1.428684   .1499367     9.53   0.000     1.134173    1.723194
------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store cr_any

.
. * C2. Reply rate. A 0.10 increase is a 10 percentage-point higher lagged reply rate.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.lag_mr_rate_centered ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month
> us100 == 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
```

```
HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  13,    557) =     100.52
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3036
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                                       (Std. err. adjusted for 558 clusters in hotel_id_num)
------------------------------------------------------------------------------------------------------------
                                           |               Robust
                       ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------------------------------------+----------------------------------------------------------------
                          sim_mean_centered |  -.1839724   .0572622    -3.21   0.001    -.2964486   -.0714961
                       lag_mr_rate_centered |    .001294   .0012058     1.07   0.284    -.0010745    .0036625
                                           |
c.sim_mean_centered#c.lag_mr_rate_centered |   .0386804   .0256797     1.51   0.133    -.0117605    .0891213
                                           |
                           ln_recent_volumn |   .0809088     .00873     9.27   0.000     .0637611    .0980566
                                  recent_sd |  -.0019205   .0075601    -0.25   0.800    -.0167703    .0129293
                              recent_rating |   .0181071   .0062717     2.89   0.004      .005788    .0304263
                          ln_lag_volumn_acc |   .0290548   .0071481     4.06   0.000     .0150143    .0430952
                         lag_avg_rating_acc |  -.0006757   .0278143    -0.02   0.981    -.0553095     .053958
                                 lag_sd_acc |  -.0170657   .0344937    -0.49   0.621    -.0848193     .050688
                       lag_avg_rating_month |   .0041845   .0017018     2.46   0.014     .0008417    .0075272
                          ln_avg_com_RevPAR |   .1209269   .0120135    10.07   0.000     .0973297    .1445242
                   ln_lag_RevPAR_clean_w199 |   .4394332   .0176578    24.89   0.000     .4047493    .4741171
                                 lag_mr_any |   .0049713    .004728     1.05   0.294    -.0043156    .0142583
                                      _cons |   1.428555   .1499094     9.53   0.000     1.134098    1.723012
------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store cr_rate

.
. * C3. Reply count.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.lag_mr_count_centered ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month
> cus100 == 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  13,    557) =     102.19
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8743
                                                  Within R-sq.    =     0.3037
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                                        (Std. err. adjusted for 558 clusters in hotel_id_num)
-------------------------------------------------------------------------------------------------------------
                                            |               Robust
                        ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
--------------------------------------------+----------------------------------------------------------------
```

```
                     sim_mean_centered |  -.1864853   .0572992    -3.25   0.001    -.2990343   -.0739363
                  lag_mr_count_centered |  -.0004081   .0002272    -1.80   0.073    -.0008543    .0000382
                                        |
c.sim_mean_centered#c.lag_mr_count_centered |  -.0022997   .0044708    -0.51   0.607    -.0110813    .0064819
                                        |
                       ln_recent_volumn |   .0839472   .0086147     9.74   0.000      .067026    .1008684
                              recent_sd |   -.001946   .0075652    -0.26   0.797    -.0168057    .0129137
                          recent_rating |   .0181352   .0062693     2.89   0.004     .0058208    .0304495
                      ln_lag_volumn_acc |   .0291072   .0071302     4.08   0.000     .0151018    .0431125
                     lag_avg_rating_acc |  -.0011808    .027843    -0.04   0.966    -.0558709    .0535093
                             lag_sd_acc |  -.0178535   .0345286    -0.52   0.605    -.0856757    .0499686
                   lag_avg_rating_month |   .0042591   .0017061     2.50   0.013      .000908    .0076102
                      ln_avg_com_RevPAR |   .1208301   .0120198    10.05   0.000     .0972205    .1444398
                 ln_lag_RevPAR_clean_w199 |   .4393448   .0176625    24.87   0.000     .4046515    .4740381
                             lag_mr_any |   .0082569    .004633     1.78   0.075    -.0008434    .0173571
                                  _cons |   1.421694   .1502004     9.47   0.000     1.126666    1.716723
------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store cr_count

.
. * C4. Total reply text length.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.ln_lag_mr_words_centered ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_mo
> _focus100 == 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  13,    557) =     102.77
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3037
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                                     (Std. err. adjusted for 558 clusters in hotel_id_num)
----------------------------------------------------------------------------------------------------------------
                                           |               Robust
                      ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------------------------------------+--------------------------------------------------------------------
                         sim_mean_centered |   -.183642    .057209    -3.21   0.001    -.2960137   -.0712703
                  ln_lag_mr_words_centered |  -.0040111   .0023402    -1.71   0.087    -.0086077    .0005855
                                           |
c.sim_mean_centered#c.ln_lag_mr_words_centered |   .0042797   .0146853     0.29   0.771    -.0245655     .033125
                                           |
                          ln_recent_volumn |   .0831727   .0085819     9.69   0.000     .0663159    .1000296
                                 recent_sd |  -.0019549   .0075755    -0.26   0.796    -.0168349    .0129251
                             recent_rating |   .0180177   .0062716     2.87   0.004     .0056988    .0303367
                         ln_lag_volumn_acc |   .0291949   .0071474     4.08   0.000     .0151557     .043234
                        lag_avg_rating_acc |  -.0008936   .0278421    -0.03   0.974    -.0555819    .0537947
                                lag_sd_acc |  -.0176329   .0345232    -0.51   0.610    -.0854444    .0501786
                      lag_avg_rating_month |   .0041678   .0017074     2.44   0.015      .000814    .0075216
                         ln_avg_com_RevPAR |   .1209439   .0120233    10.06   0.000     .0973273    .1445604
```

```
                 ln_lag_RevPAR_clean_w199 |    .439646   .0176349    24.93   0.000      .405007    .4742851
                               lag_mr_any |   .0271826   .0129616     2.10   0.036     .0017231    .0526421
                                    _cons |    1.41166   .1504329     9.38   0.000     1.116175    1.707145
------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store cr_words

.
. * C5. Average reply length.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.ln_lag_mr_avg_words_centered ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_ratir
> mple_focus100 == 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  13,    557) =     101.03
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3036
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                                               (Std. err. adjusted for 558 clusters in hotel_id_num)
----------------------------------------------------------------------------------------------------------------
                                             |               Robust
                        ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
---------------------------------------------+------------------------------------------------------------------
                           sim_mean_centered |  -.1828114   .0573081    -3.19   0.002    -.2953778    -.070245
                ln_lag_mr_avg_words_centered |   .0010693   .0052876     0.20   0.840    -.0093167    .0114554
                                             |
c.sim_mean_centered#c.ln_lag_mr_avg_words_centered |   .0099601   .0195732     0.51   0.611    -.0284861    .0484064
                                             |
                            ln_recent_volumn |   .0806417   .0087493     9.22   0.000      .063456    .0978274
                                   recent_sd |  -.0019395   .0075671    -0.26   0.798     -.016803     .012924
                               recent_rating |   .0181025   .0062772     2.88   0.004     .0057726    .0304323
                           ln_lag_volumn_acc |    .029025   .0071636     4.05   0.000      .014954    .0430959
                          lag_avg_rating_acc |  -.0005869   .0278391    -0.02   0.983    -.0552693    .0540956
                                  lag_sd_acc |  -.0170877   .0345124    -0.50   0.621     -.084878    .0507026
                        lag_avg_rating_month |   .0042387   .0017334     2.45   0.015     .0008338    .0076436
                            ln_avg_com_RevPAR |   .1209071     .01202    10.06   0.000     .0972971    .1445171
                    ln_lag_RevPAR_clean_w199 |   .4393247   .0176356    24.91   0.000     .4046843     .473965
                                  lag_mr_any |   .0018695    .022971     0.08   0.935    -.0432507    .0469898
                                       _cons |    1.43102   .1506362     9.50   0.000     1.135136    1.726904
----------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation
```

```
. est store cr_avgwords

.
. * C6. Fast replies within seven days.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.lag_mr_quick7_share_centered ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_ratir
> mple_focus100 == 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  13,     557) =     101.53
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3036
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                                               (Std. err. adjusted for 558 clusters in hotel_id_num)
---------------------------------------------------------------------------------------------------------------------
                                             |               Robust
                        ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
---------------------------------------------+-----------------------------------------------------------------------
                           sim_mean_centered |  -.1847327   .0570662    -3.24   0.001    -.2968239   -.0726414
                lag_mr_quick7_share_centered |   .0057936   .0064363     0.90   0.368    -.0068487     .018436
                                             |
c.sim_mean_centered#c.lag_mr_quick7_share_centered |  -.0241034   .0901541    -0.27   0.789    -.2011869    .1529801
                                             |
                            ln_recent_volumn |   .0806568   .0087558     9.21   0.000     .0634583    .0978553
                                   recent_sd |  -.0018612    .007565    -0.25   0.806    -.0167207    .0129983
                               recent_rating |   .0180396   .0062757     2.87   0.004     .0057126    .0303665
                           ln_lag_volumn_acc |   .0290209   .0071416     4.06   0.000     .0149931    .0430486
                          lag_avg_rating_acc |  -.0006172   .0278288    -0.02   0.982    -.0552794    .0540449
                                  lag_sd_acc |  -.0172286   .0345108    -0.50   0.618    -.0850159    .0505587
                        lag_avg_rating_month |   .0042303   .0017059     2.48   0.013     .0008795    .0075812
                           ln_avg_com_RevPAR |   .1208441   .0120057    10.07   0.000      .097262    .1444262
                     ln_lag_RevPAR_clean_w199 |   .4391942    .017644    24.89   0.000     .4045372    .4738512
                                  lag_mr_any |   .0019649   .0061667     0.32   0.750     -.010148    .0140778
                                       _cons |   1.432211   .1499114     9.55   0.000      1.13775    1.726672
---------------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store cr_quick

.
. * C7. Invite wording as engagement/solicitation language.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.lag_mr_invite_share_centered ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_ratir
> mple_focus100 == 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  13,     557) =     101.75
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
```

True!

WHY?

为什么 invite share 大
反而可以缓解？
↳ 得去回看定义！

```
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8743
                                                  Within R-sq.    =     0.3039
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2299

                                                                  (Std. err. adjusted for 558 clusters in hotel_id_num)
---------------------------------------------------------------------------------------------------------------------
                                                   |               Robust
                               ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
---------------------------------------------------+-----------------------------------------------------------------
                                  sim_mean_centered |  -.1813174   .0573961    -3.16   0.002    -.2940566   -.0685782
                      lag_mr_invite_share_centered |  -.0122619   .0085852    -1.43   0.154    -.0291252    .0046015
                                                   |
c.sim_mean_centered#c.lag_mr_invite_share_centered |   .4980094   .1797498     2.77   0.006     .1449391    .8510797
                                                   |
                                   ln_recent_volumn |   .0803952   .0086979     9.24   0.000     .0633106    .0974798
                                          recent_sd |  -.0022524   .0075839    -0.30   0.767    -.0171489     .012644
                                      recent_rating |   .0181027   .0062778     2.88   0.004     .0057717    .0304337
                                  ln_lag_volumn_acc |   .0291405   .0071506     4.08   0.000     .0150952    .0431859
                                 lag_avg_rating_acc |  -.0004584    .027805    -0.02   0.987    -.0550739    .0541572
                                         lag_sd_acc |  -.0169038   .0344524    -0.49   0.624    -.0845763    .0507687
                               lag_avg_rating_month |   .0044096   .0017002     2.59   0.010     .0010701    .0077491
                                  ln_avg_com_RevPAR |   .1209748    .011997    10.08   0.000       .09741    .1445397
                            ln_lag_RevPAR_clean_w199 |   .4392889    .017653    24.88   0.000     .4046144    .4739634
                                         lag_mr_any |   .0083094   .0044799     1.85   0.064    -.0004901     .017109
                                              _cons |   1.426673   .1498026     9.52   0.000     1.132426     1.72092
---------------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-------------------------------------------------------+
  Absorbed FE | Categories  - Redundant  = Num. Coefs |
--------------+----------------------------------------|
 hotel_id_num |       558         558           0    *|
           ym |       135           1         134     |
-------------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store cr_invite

.
. * C8. Service-recovery wording.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.lag_mr_recovery_share_centered ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rat
> sample_focus100 == 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  13,    557) =     100.61
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3036
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                                                    (Std. err. adjusted for 558 clusters in hotel_id_num)
-----------------------------------------------------------------------------------------------------------------------
                                                     |               Robust
                                 ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-----------------------------------------------------+-----------------------------------------------------------------
                                    sim_mean_centered |    -.18547   .0572462    -3.24   0.001    -.2979149   -.0730251
                      lag_mr_recovery_share_centered |   .0122357   .0097975     1.25   0.212    -.0070088    .0314802
                                                     |
```

```
c.sim_mean_centered#c.lag_mr_recovery_share_centered |  -.2208952   .2114212    -1.04   0.297    -.6361755    .1943852
                                                     |
                                     ln_recent_volumn |   .0809394   .0087512     9.25   0.000       .06375    .0981288
                                            recent_sd |  -.0019066   .0075568    -0.25   0.801    -.0167499    .0129368
                                        recent_rating |   .0181333   .0062732     2.89   0.004     .0058112    .0304553
                                    ln_lag_volumn_acc |   .0289772   .0071413     4.06   0.000     .0149501    .0430044
                                   lag_avg_rating_acc |  -.0009662    .027871    -0.03   0.972    -.0557113     .053779
                                           lag_sd_acc |  -.0172077     .03454    -0.50   0.619    -.0850522    .0506369
                                 lag_avg_rating_month |   .0044776   .0017044     2.63   0.009     .0011297    .0078255
                                    ln_avg_com_RevPAR |   .1208544   .0120092    10.06   0.000     .0972655    .1444432
                               ln_lag_RevPAR_clean_w199 |   .4391929   .0176508    24.88   0.000     .4045226    .4738632
                                           lag_mr_any |   .0050162   .0047553     1.05   0.292    -.0043243    .0143567
                                                _cons |   1.430167   .1501351     9.53   0.000     1.135267    1.725067
------------------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-------------------------------------------------------+
  Absorbed FE | Categories  - Redundant  = Num. Coefs |
--------------+----------------------------------------|
 hotel_id_num |       558         558           0    *|
           ym |       135           1         134     |
-------------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store cr_recovery

.
. * C9. Positive reply wording.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.lag_mr_positive_share_centered ln_recent_volumn recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rat
> sample_focus100 == 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  13,    557) =     101.56
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3036
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                                         (Std. err. adjusted for 558 clusters in hotel_id_num)
------------------------------------------------------------------------------------------------------------------------
                                                     |               Robust
                                 ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-----------------------------------------------------+------------------------------------------------------------------
                                    sim_mean_centered |  -.1843441    .057164    -3.22   0.001    -.2966275   -.0720607
                       lag_mr_positive_share_centered |  -.0000505   .0064932    -0.01   0.994    -.0128046    .0127036
                                                     |
c.sim_mean_centered#c.lag_mr_positive_share_centered |   .0240148   .1038576     0.23   0.817    -.1799855    .2280152
                                                     |
                                     ln_recent_volumn |   .0807568   .0087499     9.23   0.000       .06357    .0979436
                                            recent_sd |  -.0018703   .0075464    -0.25   0.804    -.0166932    .0129526
                                        recent_rating |   .0180951   .0062702     2.89   0.004      .005779    .0304112
                                    ln_lag_volumn_acc |   .0290457   .0071447     4.07   0.000     .0150119    .0430795
                                   lag_avg_rating_acc |  -.0007008   .0278315    -0.03   0.980    -.0553684    .0539668
                                           lag_sd_acc |  -.0172269   .0345097    -0.50   0.618    -.0850119    .0505582
                                 lag_avg_rating_month |   .0042334   .0017332     2.44   0.015      .000829    .0076378
                                    ln_avg_com_RevPAR |   .1208702   .0120226    10.05   0.000     .0972552    .1444853
                               ln_lag_RevPAR_clean_w199 |    .439285   .0176374    24.91   0.000     .4046411    .4739289
                                           lag_mr_any |   .0062064   .0057432     1.08   0.280    -.0050745    .0174874
                                                _cons |    1.42933   .1501071     9.52   0.000     1.134485    1.724176
```

```
------------------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-------------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-------------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store cr_positive

.
. esttab cr_any cr_rate cr_count cr_words cr_avgwords cr_quick cr_invite cr_recovery cr_positive using "`table_dir'/story_table_c_reply_revenue_260524.rtf", replace star(* 0.10 ** 0.05
> , labels("Observations" "Adjusted R-squared")) mtitles("any reply" "reply rate" "reply count" "words" "avg words" "quick7" "invite" "recovery" "positive") nogap compress
(output written to /Users/samxie/Research/ReviewSimi_Sales/Code/outputs/core_simi_260501/tables_explicit/story_table_c_reply_revenue_260524.rtf)

. esttab cr_any cr_rate cr_count cr_words cr_avgwords cr_quick cr_invite cr_recovery cr_positive using "`csv_dir'/story_table_c_reply_revenue_260524.csv", replace csv star(* 0.10 ** 0.0
> _a, labels("Observations" "Adjusted R-squared")) mtitles("any reply" "reply rate" "reply count" "words" "avg words" "quick7" "invite" "recovery" "positive") nogap
(output written to /Users/samxie/Research/ReviewSimi_Sales/Code/outputs/core_simi_260501/csv/story_table_c_reply_revenue_260524.csv)


.
. *******************************************************
. ************ 4.1 Route C: text interactions ***********
. *******************************************************
```

⇒ 更Solid做法：

训自己的ml来打分！

/label后打分即可

```
.
. estimates clear

.
. * C10. Thanks wording.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.lag_mr_thanks_share_centered recent_sd recent_rating ln_recent_volumn ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_ratir
> mple_focus100 == 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  13,    557) =     103.74
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8743
                                                  Within R-sq.    =     0.3038
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2299

                                                              (Std. err. adjusted for 558 clusters in hotel_id_num)
---------------------------------------------------------------------------------------------------------------------
                                                    |               Robust
                               ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
----------------------------------------------------+----------------------------------------------------------------
                                  sim_mean_centered |  -.1810791   .0571203    -3.17   0.002    -.2932766   -.0688817
                       lag_mr_thanks_share_centered |   .0298153   .0113476     2.63   0.009      .007526    .0521045
                                                    |
c.sim_mean_centered#c.lag_mr_thanks_share_centered |   .0347288   .0863035     0.40   0.688    -.1347914    .2042489
                                                    |
                                          recent_sd |  -.0014473   .0075581    -0.19   0.848    -.0162933    .0133986
                                      recent_rating |   .0181168   .0062681     2.89   0.004     .0058048    .0304287
                                   ln_recent_volumn |   .0806828   .0087567     9.21   0.000     .0634827    .0978829
                                  ln_lag_volumn_acc |   .0290715   .0071613     4.06   0.000      .015005     .043138
                                 lag_avg_rating_acc |  -.0008221   .0278274    -0.03   0.976    -.0554817    .0538374
                                         lag_sd_acc |   -.017165   .0344968    -0.50   0.619    -.0849248    .0505947
                               lag_avg_rating_month |   .0039539     .00171     2.31   0.021      .000595    .0073128
                                   ln_avg_com_RevPAR |   .1208503   .0120081    10.06   0.000     .0972635    .1444371
```

```
                  ln_lag_RevPAR_clean_w199 |   .4391478   .0175911    24.96   0.000     .4045948    .4737007
                                lag_mr_any |  -.0207523   .0115184    -1.80   0.072     -.043377    .0018724
                                     _cons |   1.444232    .148898     9.70   0.000     1.151762    1.736702
-----------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store tc_thanks

.
. * C11. Apology wording.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.lag_mr_apology_share_centered recent_sd recent_rating ln_recent_volumn ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rati
> ample_focus100 == 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  13,    557) =     100.38
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3036
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                                            (Std. err. adjusted for 558 clusters in hotel_id_num)
-----------------------------------------------------------------------------------------------------------------
                                                    |               Robust
                               ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
----------------------------------------------------+------------------------------------------------------------
                                  sim_mean_centered |  -.1879007    .057139    -3.29   0.001    -.3001349   -.0756665
                      lag_mr_apology_share_centered |  -.0009837   .0079154    -0.12   0.901    -.0165314    .0145641
                                                    |
c.sim_mean_centered#c.lag_mr_apology_share_centered |   -.215413   .1602511    -1.34   0.179    -.5301833    .0993573
                                                    |
                                          recent_sd |  -.0016136     .00758    -0.21   0.832    -.0165025    .0132753
                                      recent_rating |   .0180791   .0062709     2.88   0.004     .0057615    .0303967
                                   ln_recent_volumn |   .0809429   .0087352     9.27   0.000     .0637849    .0981009
                                  ln_lag_volumn_acc |   .0290786    .007153     4.07   0.000     .0150284    .0431288
                                 lag_avg_rating_acc |  -.0008304   .0278308    -0.03   0.976    -.0554966    .0538357
                                         lag_sd_acc |  -.0173656   .0345273    -0.50   0.615    -.0851852     .050454
                               lag_avg_rating_month |   .0043309   .0017475     2.48   0.013     .0008984    .0077634
                                  ln_avg_com_RevPAR |   .1208577   .0120071    10.07   0.000      .097273    .1444424
                           ln_lag_RevPAR_clean_w199 |   .4392199   .0176543    24.88   0.000     .4045428     .473897
                                         lag_mr_any |   .0059589   .0049053     1.21   0.225    -.0036763    .0155941
                                              _cons |   1.429003   .1499207     9.53   0.000     1.134524    1.723482
-----------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation
```

```
. est store tc_apology

.
. * C12. Contact wording.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.lag_mr_contact_share_centered recent_sd recent_rating ln_recent_volumn ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rati
> ample_focus100 == 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  13,    557) =     101.14
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3036
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                                                 (Std. err. adjusted for 558 clusters in hotel_id_num)
-----------------------------------------------------------------------------------------------------------------------
                                                |               Robust
                           ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
------------------------------------------------+----------------------------------------------------------------------
                              sim_mean_centered |   -.184996   .0570268    -3.24   0.001    -.2970098   -.0729821
                  lag_mr_contact_share_centered |   .0026498     .00669     0.40   0.692    -.0104908    .0157905
                                                |
c.sim_mean_centered#c.lag_mr_contact_share_centered |  -.0706725   .0969239    -0.73   0.466    -.2610536    .1197086
                                                |
                                      recent_sd |  -.0018531   .0075599    -0.25   0.806    -.0167025    .0129963
                                  recent_rating |   .0181133   .0062682     2.89   0.004      .005801    .0304255
                               ln_recent_volumn |   .0809468   .0087523     9.25   0.000     .0637552    .0981385
                              ln_lag_volumn_acc |    .029051   .0071542     4.06   0.000     .0149986    .0431035
                             lag_avg_rating_acc |  -.0007512   .0278271    -0.03   0.978    -.0554101    .0539078
                                     lag_sd_acc |  -.0171636   .0344953    -0.50   0.619    -.0849204    .0505931
                           lag_avg_rating_month |   .0043181   .0017205     2.51   0.012     .0009386    .0076976
                               ln_avg_com_RevPAR |   .1208412   .0120137    10.06   0.000     .0972435     .144439
                       ln_lag_RevPAR_clean_w199 |   .4392147   .0176417    24.90   0.000     .4045624    .4738671
                                     lag_mr_any |   .0050933   .0055806     0.91   0.362    -.0058682    .0160548
                                          _cons |   1.429433   .1499195     9.53   0.000     1.134956     1.72391
-----------------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store tc_contact

.
. * C13. Personalization.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.lag_mr_personal_share_centered recent_sd recent_rating ln_recent_volumn ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rat
> sample_focus100 == 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  13,    557) =     100.86
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
```

```
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3036
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                                                   (Std. err. adjusted for 558 clusters in hotel_id_num)
-----------------------------------------------------------------------------------------------------------------------
                                                  |               Robust
                               ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
----------------------------------------------------+------------------------------------------------------------------
                                   sim_mean_centered |  -.1832409   .0572478    -3.20   0.001    -.2956889   -.0707929
                        lag_mr_personal_share_centered |  -.0027332   .0263694    -0.10   0.917    -.0545288    .0490623
                                                    |
c.sim_mean_centered#c.lag_mr_personal_share_centered |   .0444345   .0824541     0.54   0.590    -.1175245    .2063936
                                                    |
                                           recent_sd |  -.0019262   .0075737    -0.25   0.799    -.0168027    .0129504
                                       recent_rating |   .0180889   .0062803     2.88   0.004      .005753    .0304248
                                   ln_recent_volumn |   .0806477   .0087511     9.22   0.000     .0634585    .0978368
                                  ln_lag_volumn_acc |   .0290574   .0071487     4.06   0.000     .0150156    .0430991
                                 lag_avg_rating_acc |  -.0005933   .0278424    -0.02   0.983    -.0552823    .0540957
                                         lag_sd_acc |  -.0170688   .0345212    -0.49   0.621    -.0848764    .0507388
                               lag_avg_rating_month |   .0042013   .0017046     2.46   0.014     .0008529    .0075496
                                  ln_avg_com_RevPAR |   .1209134   .0120206    10.06   0.000     .0973021    .1445248
                              ln_lag_RevPAR_clean_w199 |   .4393336   .0176404    24.90   0.000     .4046837    .4739835
                                         lag_mr_any |   .0089882   .0265639     0.34   0.735    -.0431896    .0611659
                                              _cons |   1.427446   .1503204     9.50   0.000     1.132182     1.72271
-----------------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store tc_personal

.
. * C14. Negative-problem wording.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean##c.lag_mr_negtone_share recent_sd recent_rating ln_recent_volumn ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg_co
> 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  13,    557) =     100.50
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3037
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                   (Std. err. adjusted for 558 clusters in hotel_id_num)
---------------------------------------------------------------------------------------
                                  |               Robust
            ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
----------------------------------+----------------------------------------------------
                         sim_mean |  -.1725925   .0583048    -2.96   0.003    -.2871167   -.0580684
             lag_mr_negtone_share |   .0800435   .0536989     1.49   0.137    -.0254336    .1855207
                                  |
```

这具体是什么？

```
c.sim_mean#c.lag_mr_negtone_share |  -.3198647   .1873077    -1.71   0.088    -.6877804     .048051
                                  |
                        recent_sd |  -.0016513   .0075628    -0.22   0.827    -.0165064    .0132037
                    recent_rating |    .018043   .0062704     2.88   0.004     .0057264    .0303595
                  ln_recent_volumn |   .0809612    .008739     9.26   0.000     .0637957    .0981266
                ln_lag_volumn_acc |   .0290467   .0071418     4.07   0.000     .0150184    .0430749
                lag_avg_rating_acc |  -.0007531   .0278411    -0.03   0.978    -.0554394    .0539332
                       lag_sd_acc |  -.0173539   .0345318    -0.50   0.615    -.0851824    .0504746
              lag_avg_rating_month |   .0040021   .0017277     2.32   0.021     .0006085    .0073956
                 ln_avg_com_RevPAR |   .1207963   .0120112    10.06   0.000     .0972035    .1443891
             ln_lag_RevPAR_clean_w199 |   .4392265   .0176502    24.89   0.000     .4045573    .4738956
                       lag_mr_any |    .007461   .0048581     1.54   0.125    -.0020815    .0170035
                            _cons |   1.481157   .1501005     9.87   0.000     1.186325     1.77599
----------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store tc_negtone

.
```

模板化又是怎么得到的？

```
. * C15. Template-like wording.
. reghdfe ln_RevPAR_clean_w199 c.sim_mean_centered##c.lag_mr_template_share_centered recent_sd recent_rating ln_recent_volumn ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rat
> sample_focus100 == 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  13,    557) =     103.03
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8742
                                                  Within R-sq.    =     0.3036
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2300

                                        (Std. err. adjusted for 558 clusters in hotel_id_num)
-----------------------------------------------------------------------------------------------------
                                               |               Robust
                          ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-----------------------------------------------+-----------------------------------------------------
                             sim_mean_centered |  -.1848872   .0572056    -3.23   0.001    -.2972523   -.0725222
                 lag_mr_template_share_centered |  -.0081349   .0204286    -0.40   0.691    -.0482614    .0319915
                                               |
c.sim_mean_centered#c.lag_mr_template_share_centered |  -.0913917   .2689001    -0.34   0.734    -.6195739    .4367905
                                               |
                                     recent_sd |  -.0019376   .0075676    -0.26   0.798     -.016802    .0129269
                                 recent_rating |    .018154   .0062584     2.90   0.004     .0058611    .0304469
                              ln_recent_volumn |   .0809154   .0087747     9.22   0.000     .0636799    .0981508
                             ln_lag_volumn_acc |    .029182   .0071206     4.10   0.000     .0151954    .0431686
                            lag_avg_rating_acc |  -.0003705   .0278364    -0.01   0.989    -.0550476    .0543066
                                    lag_sd_acc |  -.0167855   .0343341    -0.49   0.625    -.0842256    .0506546
                          lag_avg_rating_month |   .0042448    .001709     2.48   0.013     .0008878    .0076017
                             ln_avg_com_RevPAR |   .1208474    .012024    10.05   0.000     .0972295    .1444654
                          ln_lag_RevPAR_clean_w199 |   .4392088    .017675    24.85   0.000      .404491    .4739265
                                    lag_mr_any |    .006693   .0046832     1.43   0.154     -.002506    .0158919
                                         _cons |   1.426382   .1493908     9.55   0.000     1.132943     1.71982
```

```
------------------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
           ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store tc_template

.
. * C16. Three-way test: volume x ARS x average reply length.
. reghdfe ln_RevPAR_clean_w199 c.ln_recent_volumn_centered##c.sim_mean##c.ln_lag_mr_avg_words_centered recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_ra
> _sample_focus100 == 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  16,     557) =      87.78
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8743
                                                  Within R-sq.    =     0.3040
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2299

                                                                  (Std. err. adjusted for 558 clusters in hotel_id_num)
---------------------------------------------------------------------------------------------------------------------------
                                                                   |               Robust
                                              ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------------------------------------------------------------+-------------------------------------------------------------------
                                         ln_recent_volumn_centered |   .1704117   .0466797     3.65   0.000     .0787219    .2621015
                                                          sim_mean |  -.1818128   .0575251    -3.16   0.002    -.2948055   -.0688202
                                                                   |
                             c.ln_recent_volumn_centered#c.sim_mean |  -.2829756   .1468955    -1.93   0.055    -.5715124    .0055611
                                                                   |
                                      ln_lag_mr_avg_words_centered |  -.0041899   .0075871    -0.55   0.581    -.0190927    .0107129
                                                                   |
         c.ln_recent_volumn_centered#c.ln_lag_mr_avg_words_centered |   .0247927   .0119871     2.07   0.039     .0012473    .0483382
                                                                   |
                         c.sim_mean#c.ln_lag_mr_avg_words_centered |    .020655   .0200377     1.03   0.303    -.0187037    .0600136
                                                                   |
c.ln_recent_volumn_centered#c.sim_mean#c.ln_lag_mr_avg_words_centered |  -.0939866   .0386473    -2.43   0.015    -.1698989   -.0180744
                                                                   |
```

Reply 越多越不好!

```
                                                         recent_sd |   -.001993   .0075569    -0.26   0.792    -.0168365    .0128505
                                                     recent_rating |   .0179727   .0062718     2.87   0.004     .0056534     .030292
                                                 ln_lag_volumn_acc |   .0287943   .0072451     3.97   0.000     .0145632    .0430254
                                                lag_avg_rating_acc |  -.0027849    .027966    -0.10   0.921    -.0577165    .0521468
                                                        lag_sd_acc |  -.0193043   .0347094    -0.56   0.578    -.0874816    .0488729
                                              lag_avg_rating_month |   .0041817   .0017416     2.40   0.017     .0007607    .0076026
                                                 ln_avg_com_RevPAR |   .1206061   .0120157    10.04   0.000     .0970045    .1442076
                                              ln_lag_RevPAR_clean_w199 |   .4388439   .0176825    24.82   0.000     .4041113    .4735764
                                                        lag_mr_any |  -.0007945   .0231155    -0.03   0.973    -.0461988    .0446097
                                                             _cons |   1.722516    .155471    11.08   0.000     1.417135    2.027897
---------------------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
```

```
 hotel_id_num |       558         558           0    *|
           ym |       135           1         134     |
-------------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store tc_triple_avgw

.
. * C17. Three-way test: volume x ARS x quick reply.
. reghdfe ln_RevPAR_clean_w199 c.ln_recent_volumn_centered##c.sim_mean_centered##c.lag_mr_quick7_share_centered recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc l
> f cs_sample_focus100 == 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  16,     557) =      59.70
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8733
                                                  Adj R-squared   =     0.8705
                                                  Within R-sq.    =     0.2831
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2333

                                                                                           (Std. err. adjusted for 558 clusters in hotel_id_num)
------------------------------------------------------------------------------------------------------------------------------------------------
                                                                             |               Robust
                                                        ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-----------------------------------------------------------------------------+------------------------------------------------------------------
                                                    ln_recent_volumn_centered |   .0941153   .0094237     9.99   0.000     .0756049    .1126256
                                                           sim_mean_centered |  -.1899645   .0607979    -3.12   0.002    -.3093857   -.0705433
                                                                             |
```

削弱recent volume

```
                                   c.ln_recent_volumn_centered#c.sim_mean_centered |  -.2938732   .1559418    -1.88   0.060    -.6001791    .0124327
                                                                             |
                                               lag_mr_quick7_share_centered |   .0086512   .0067625     1.28   0.201    -.0046318    .0219343
                                                                             |
                c.ln_recent_volumn_centered#c.lag_mr_quick7_share_centered |  -.0075542   .0140641    -0.54   0.591    -.0351794     .020071
                                                                             |
                        c.sim_mean_centered#c.lag_mr_quick7_share_centered |   .0028748   .0996637     0.03   0.977    -.1928878    .1986375
                                                                             |
c.ln_recent_volumn_centered#c.sim_mean_centered#c.lag_mr_quick7_share_centered |  -.4451224    .206678    -2.15   0.032    -.8510859   -.0391588
                                                                             |
```

不好解释了!

```
                                                                   recent_sd |  -.0012078   .0081159    -0.15   0.882    -.0171493    .0147338
                                                               recent_rating |   .0189032   .0068345     2.77   0.006     .0054786    .0323278
                                                           ln_lag_volumn_acc |   .0360368   .0079829     4.51   0.000     .0203566     .051717
                                                          lag_avg_rating_acc |  -.0054393   .0307218    -0.18   0.860    -.0657841    .0549055
                                                                  lag_sd_acc |  -.0207156    .038169    -0.54   0.588    -.0956884    .0542572
                                                        lag_avg_rating_month |   .0033962   .0018712     1.82   0.070    -.0002792    .0070716
                                                            ln_avg_com_RevPAR |   .1266506   .0128898     9.83   0.000     .1013319    .1519692
                                                           ln_lag_RevPAR_clean |   .3739681   .0226702    16.50   0.000     .3294386    .4184976
                                                                  lag_mr_any |    .000921   .0064025     0.14   0.886    -.0116549    .0134969
                                                                       _cons |   1.889018   .1715554    11.01   0.000     1.552043    2.225992
------------------------------------------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-------------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+-----------------------------------------|
 hotel_id_num |       558         558           0    *|
           ym |       135           1         134     |
-------------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store tc_triple_quick
```

```
.
. * C18. Three-way test: volume x ARS x positive reply wording.
. reghdfe ln_RevPAR_clean_w199 c.ln_recent_volumn_centered##c.sim_mean_centered##c.lag_mr_positive_share_centered recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc
> r_any if cs_sample_focus100 == 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  16,     557) =      86.08
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8743
                                                  Within R-sq.    =     0.3039
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2299

                                                                                         (Std. err. adjusted for 558 clusters in hotel_id_num)
------------------------------------------------------------------------------------------------------------------------------------------------
                                                                         |               Robust
                                                    ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------------------------------------------------------------------+----------------------------------------------------------------------
                                               ln_recent_volumn_centered |   .0846261    .009234     9.16   0.000     .0664883    .1027639
                                                       sim_mean_centered |  -.1804787   .0580912    -3.11   0.002    -.2945832   -.0663742
                                                                         |
                             c.ln_recent_volumn_centered#c.sim_mean_centered |   -.254683   .1503891    -1.69   0.091     -.550082    .0407161
                                                                         |
                                           lag_mr_positive_share_centered |   .0017305   .0066748     0.26   0.796    -.0113803    .0148413
                                                                         |
                c.ln_recent_volumn_centered#c.lag_mr_positive_share_centered |   .0102737   .0130477     0.79   0.431     -.015355    .0359024
                                                                         |
                        c.sim_mean_centered#c.lag_mr_positive_share_centered |   .0369652    .110164     0.34   0.737    -.1794224    .2533528
                                                                         |
c.ln_recent_volumn_centered#c.sim_mean_centered#c.lag_mr_positive_share_centered |  -.5677855   .2416443    -2.35   0.019    -1.042431   -.0931399
                                                                         |
                                                               recent_sd |  -.0017844   .0075301    -0.24   0.813    -.0165753    .0130064
                                                           recent_rating |   .0179813   .0062675     2.87   0.004     .0056705    .0302921
                                                       ln_lag_volumn_acc |   .0294215   .0071332     4.12   0.000     .0154102    .0434328
                                                      lag_avg_rating_acc |  -.0017896   .0279112    -0.06   0.949    -.0566137    .0530346
                                                              lag_sd_acc |  -.0180568   .0346837    -0.52   0.603    -.0861836    .0500701
                                                    lag_avg_rating_month |    .004238   .0017388     2.44   0.015     .0008225    .0076535
                                                       ln_avg_com_RevPAR |   .1206415   .0120216    10.04   0.000     .0970283    .1442548
                                                    ln_lag_RevPAR_clean_w199 |   .4387323   .0176665    24.83   0.000     .4040312    .4734334
                                                              lag_mr_any |   .0058332   .0057165     1.02   0.308    -.0053954    .0170617
                                                                   _cons |   1.655999   .1540791    10.75   0.000     1.353352    1.958646
------------------------------------------------------------------------------------------------------------------------------------------------
```

越多好评但更相似，则越差

不足,没好的reply

```
Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store tc_triple_pos

.
. * C19. Three-way test: volume x ARS x recovery wording.
. reghdfe ln_RevPAR_clean c.ln_recent_volumn_centered##c.sim_mean_centered##c.lag_mr_recovery_share_centered recent_sd recent_rating ln_lag_volumn_acc lag_avg_rating_acc lag_sd_acc lag_
>  if cs_sample_focus100 == 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
```

```
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  16,    557) =      68.75
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8587
                                                  Adj R-squared   =     0.8557
                                                  Within R-sq.    =     0.2615
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2637

                                                                                           (Std. err. adjusted for 558 clusters in hotel_id_num)
-----------------------------------------------------------------------------------------------------------------------------------------------
                                                                               |               Robust
                                                                ln_RevPAR_clean | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
--------------------------------------------------------------------------------+--------------------------------------------------------------
                                                     ln_recent_volumn_centered |   .0822499   .0100271     8.20   0.000     .0625544    .1019455
                                                             sim_mean_centered |  -.2209818   .0667066    -3.31   0.001     -.352009   -.0899546
                                                                               |
                                   c.ln_recent_volumn_centered#c.sim_mean_centered |  -.2745595   .1606499    -1.71   0.088    -.5901132    .0409942
                                                                               |
                                              lag_mr_recovery_share_centered |   .0147246   .0111486     1.32   0.187    -.0071739    .0366231
                                                                               |
                 c.ln_recent_volumn_centered#c.lag_mr_recovery_share_centered |   .0271357    .044905     0.60   0.546    -.0610682    .1153395
                                                                               |
                         c.sim_mean_centered#c.lag_mr_recovery_share_centered |  -.1651768   .3092158    -0.53   0.593    -.7725484    .4421949
                                                                               |
c.ln_recent_volumn_centered#c.sim_mean_centered#c.lag_mr_recovery_share_centered |  -.7312391   .7854572    -0.93   0.352    -2.274059     .811581
                                                                               |
                                                                     recent_sd |  -.0057306   .0086732    -0.66   0.509    -.0227667    .0113055
                                                                 recent_rating |   .0223287    .006808     3.28   0.001     .0089563     .035701
                                                             ln_lag_volumn_acc |   .0290227   .0076205     3.81   0.000     .0140544    .0439911
                                                            lag_avg_rating_acc |  -.0003894   .0294897    -0.01   0.989     -.058314    .0575352
                                                                    lag_sd_acc |  -.0174569   .0354644    -0.49   0.623    -.0871172    .0522034
                                                          lag_avg_rating_month |   .0042098    .002041     2.06   0.040     .0002009    .0082187
                                                             ln_avg_com_RevPAR |   .1250274   .0128832     9.70   0.000     .0997218    .1503329
                                                      ln_lag_RevPAR_clean_w199 |   .4540621    .021739    20.89   0.000     .4113617    .4967625
                                                                    lag_mr_any |   .0025367   .0051692     0.49   0.624    -.0076168    .0126903
                                                                         _cons |   1.557534    .160454     9.71   0.000     1.242365    1.872703
-----------------------------------------------------------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
  Absorbed FE | Categories  - Redundant  = Num. Coefs |
--------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
           ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store tc_triple_rec

.
. esttab tc_thanks tc_apology tc_contact tc_personal tc_negtone tc_template tc_triple_avgw tc_triple_quick tc_triple_pos tc_triple_rec using "`table_dir'/story_table_c_mr_text_260524.rt
> 4)) se(par fmt(4))) stats(N r2_a, labels("Observations" "Adjusted R-squared")) mtitles("thanks" "apology" "contact" "personal" "neg tone" "template" "triple avg words" "triple quick7"
(output written to /Users/samxie/Research/ReviewSimi_Sales/Code/outputs/core_simi_260501/tables_explicit/story_table_c_mr_text_260524.rtf)

. esttab tc_thanks tc_apology tc_contact tc_personal tc_negtone tc_template tc_triple_avgw tc_triple_quick tc_triple_pos tc_triple_rec using "`csv_dir'/story_table_c_mr_text_260524.csv"
> t(4)) se(par fmt(4))) stats(N r2_a, labels("Observations" "Adjusted R-squared")) mtitles("thanks" "apology" "contact" "personal" "neg tone" "template" "triple avg words" "triple quick
(output written to /Users/samxie/Research/ReviewSimi_Sales/Code/outputs/core_simi_260501/csv/story_table_c_mr_text_260524.csv)

.
. *******************************************************
```

```
. ************ 5. Route C: mechanisms *******************
. *******************************************************
.
. estimates clear

.
. * C20. Mechanism DV: next-month review volume. MR text is treated as observable engagement, not causal solicitation.
. reghdfe ln_recent_volumn lag_mr_rate ln_lag_mr_words lag_mr_count lag_mr_quick7_share lag_mr_invite_share lag_mr_apology_share lag_mr_recovery_share lag_mr_positive_share lag_mr_templ
>  lag_sd_acc lag_avg_rating_month ln_avg_com_RevPAR ln_lag_RevPAR_clean_w199 if cs_sample_focus100 == 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,589
Absorbing 2 HDFE groups                           F(  17,    557) =      30.81
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.7017
                                                  Adj R-squared   =     0.6953
                                                  Within R-sq.    =     0.1320
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2077

                                     (Std. err. adjusted for 558 clusters in hotel_id_num)
------------------------------------------------------------------------------------------
                         |               Robust
        ln_recent_volumn | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------------------+----------------------------------------------------------------
             lag_mr_rate |  -.0281982   .0064948    -4.34   0.000    -.0409554    -.015441
         ln_lag_mr_words |   .0083709   .0028045     2.98   0.003     .0028621    .0138796
            lag_mr_count |     .00878   .0013088     6.71   0.000     .0062093    .0113508
     lag_mr_quick7_share |   .0104719   .0076269     1.37   0.170     -.004509    .0254528
     lag_mr_invite_share |  -.0116286   .0090974    -1.28   0.202    -.0294981    .0062409
    lag_mr_apology_share |  -.0041556   .0100371    -0.41   0.679    -.0238708    .0155596
   lag_mr_recovery_share |   .0009068   .0092539     0.10   0.922      -.01727    .0190836
   lag_mr_positive_share |   -.017461   .0074881    -2.33   0.020    -.0321694   -.0027526
   lag_mr_template_share |  -.0230122   .0189214    -1.22   0.224    -.0601782    .0141538
               recent_sd |  -.0373418   .0090862    -4.11   0.000    -.0551891   -.0194945
           recent_rating |      .0581   .0073717     7.88   0.000     .0436203    .0725798
       ln_lag_volumn_acc |   .0767356   .0116945     6.56   0.000     .0537649    .0997063
      lag_avg_rating_acc |   .0612123   .0332047     1.84   0.066    -.0040095    .1264341
              lag_sd_acc |   .0529813    .044552     1.19   0.235    -.0345292    .1404918
    lag_avg_rating_month |  -.0019483   .0012406    -1.57   0.117    -.0043852    .0004886
       ln_avg_com_RevPAR |   .0143362   .0070271     2.04   0.042     .0005334    .0281391
ln_lag_RevPAR_clean_w199 |   .0529739    .007348     7.21   0.000     .0385407    .0674071
                   _cons |   1.495105   .1626882     9.19   0.000     1.175548    1.814663
------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store tm_volume

.
. * C21. Mechanism DV: ARS. This asks whether reply engagement predicts later review similarity.
. reghdfe sim_mean lag_mr_rate ln_lag_mr_words lag_mr_count lag_mr_quick7_share lag_mr_invite_share lag_mr_apology_share lag_mr_recovery_share lag_mr_positive_share lag_mr_template_shar
> ating_acc lag_sd_acc lag_avg_rating_month ln_avg_com_RevPAR ln_lag_RevPAR_clean_w199 if cs_sample_focus100 == 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)
```

```
HDFE Linear regression                            Number of obs   =     33,589
Absorbing 2 HDFE groups                           F(  18,    557) =      85.20
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.5742
                                                  Adj R-squared   =     0.5651
                                                  Within R-sq.    =     0.2635
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.0292

                                     (Std. err. adjusted for 558 clusters in hotel_id_num)
------------------------------------------------------------------------------------------
                         |               Robust
                sim_mean | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------------------+----------------------------------------------------------------
             lag_mr_rate |  -.0000781   .0001974    -0.40   0.692    -.0004657    .0003095
         ln_lag_mr_words |   .0004037   .0001999     2.02   0.044      .000011    .0007964
            lag_mr_count |  -.0000481   .0000401    -1.20   0.231    -.0001267    .0000306
     lag_mr_quick7_share |  -.0013738   .0008983    -1.53   0.127    -.0031383    .0003907
     lag_mr_invite_share |  -.0006979   .0010528    -0.66   0.508    -.0027658      .00137
    lag_mr_apology_share |   -.001696   .0009775    -1.74   0.083    -.0036159     .000224
   lag_mr_recovery_share |  -.0034901    .001396    -2.50   0.013    -.0062323    -.000748
   lag_mr_positive_share |  -.0004262   .0008759    -0.49   0.627    -.0021468    .0012943
   lag_mr_template_share |   .0017605   .0021211     0.83   0.407    -.0024058    .0059268
               recent_sd |  -.0418214   .0015074   -27.74   0.000    -.0447823   -.0388604
           recent_rating |   .0131568   .0011903    11.05   0.000     .0108187    .0154948
        ln_recent_volumn |   .0204134   .0015147    13.48   0.000     .0174381    .0233886
       ln_lag_volumn_acc |   .0053208   .0011899     4.47   0.000     .0029836    .0076579
      lag_avg_rating_acc |  -.0084377   .0042079    -2.01   0.045    -.0167029   -.0001725
              lag_sd_acc |   .0138829   .0050875     2.73   0.007     .0038898     .023876
    lag_avg_rating_month |   .0001918   .0001894     1.01   0.312    -.0001802    .0005638
       ln_avg_com_RevPAR |   .0004773   .0008668     0.55   0.582    -.0012252    .0021799
ln_lag_RevPAR_clean_w199 |  -.0022581    .000931    -2.43   0.016    -.0040869   -.0004293
                   _cons |   .2290493   .0205837    11.13   0.000     .1886182    .2694804
------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store tm_ars

.
. * C22. Mechanism DV: RevPAR with MR text variables entered directly.
. reghdfe ln_RevPAR_clean_w199 lag_mr_rate ln_lag_mr_words lag_mr_count lag_mr_quick7_share lag_mr_invite_share lag_mr_apology_share lag_mr_recovery_share lag_mr_positive_share lag_mr_t
> _volumn_acc lag_avg_rating_acc lag_sd_acc lag_avg_rating_month ln_avg_com_RevPAR ln_lag_RevPAR_clean_w199 if cs_sample_focus100 == 1, absorb(hotel_id_num ym) vce(cluster hotel_id_num)
(dropped 2 singleton observations)
(MWFE estimator converged in 7 iterations)

HDFE Linear regression                            Number of obs   =     33,546
Absorbing 2 HDFE groups                           F(  19,    557) =      73.38
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.8769
                                                  Adj R-squared   =     0.8743
                                                  Within R-sq.    =     0.3039
Number of clusters (hotel_id_num) =        558    Root MSE        =     0.2299

                                     (Std. err. adjusted for 558 clusters in hotel_id_num)
```

```
----------------------------------------------------------------------------------------------
                          |               Robust
      ln_RevPAR_clean_w199 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
--------------------------+-------------------------------------------------------------------
               lag_mr_rate |    .002645   .0015862     1.67   0.096    -.0004707    .0057606
           ln_lag_mr_words |  -.0000537   .0014521    -0.04   0.971     -.002906    .0027985
              lag_mr_count |  -.0005446   .0002464    -2.21   0.027    -.0010286   -.0000606
       lag_mr_quick7_share |   .0082234   .0064859     1.27   0.205    -.0045164    .0209632
       lag_mr_invite_share |  -.0128285   .0086921    -1.48   0.141    -.0299017    .0042447
      lag_mr_apology_share |    -.00087   .0079393    -0.11   0.913    -.0164646    .0147246
     lag_mr_recovery_share |   .0141862   .0096197     1.47   0.141    -.0047092    .0330816
     lag_mr_positive_share |   .0029159   .0067767     0.43   0.667    -.0103951    .0162269
     lag_mr_template_share |  -.0069116   .0204094    -0.34   0.735    -.0470004    .0331772
          ln_recent_volumn |   .0847396    .008536     9.93   0.000     .0679728    .1015064
                  sim_mean |  -.1834357   .0571935    -3.21   0.001     -.295777   -.0710944
                 recent_sd |  -.0020392   .0075608    -0.27   0.787    -.0168903    .0128119
             recent_rating |   .0181047   .0062603     2.89   0.004     .0058079    .0304014
          ln_lag_volumn_acc |   .0290553    .007121     4.08   0.000     .0150679    .0430426
         lag_avg_rating_acc |  -.0010376   .0278433    -0.04   0.970    -.0557283    .0536531
                lag_sd_acc |  -.0179829   .0344024    -0.52   0.601    -.0855572    .0495913
      lag_avg_rating_month |   .0044628   .0017641     2.53   0.012     .0009976    .0079279
          ln_avg_com_RevPAR |   .1208922    .012016    10.06   0.000       .09729    .1444945
    ln_lag_RevPAR_clean_w199 |   .4394735   .0176819    24.85   0.000      .404742    .4742049
                     _cons |   1.473678   .1496829     9.85   0.000     1.179666    1.767689
----------------------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
 hotel_id_num |       558         558           0    *|
          ym |       135           1         134     |
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

. est store tm_revpar

.
. esttab tm_volume tm_ars tm_revpar using "`table_dir'/story_table_c_mr_mechanisms_260524.rtf", replace star(* 0.10 ** 0.05 *** 0.01 **** 0.001) cells(b(star fmt(4)) se(par fmt(4))) sta
> olume" "DV: ARS" "DV: RevPAR") nogap compress
(output written to /Users/samxie/Research/ReviewSimi_Sales/Code/outputs/core_simi_260501/tables_explicit/story_table_c_mr_mechanisms_260524.rtf)

. esttab tm_volume tm_ars tm_revpar using "`csv_dir'/story_table_c_mr_mechanisms_260524.csv", replace csv star(* 0.10 ** 0.05 *** 0.01 **** 0.001) cells(b(star fmt(4)) se(par fmt(4))) s
>  volume" "DV: ARS" "DV: RevPAR") nogap
(output written to /Users/samxie/Research/ReviewSimi_Sales/Code/outputs/core_simi_260501/csv/story_table_c_mr_mechanisms_260524.csv)

.
. log close
      name:  <unnamed>
       log:  /Users/samxie/Research/ReviewSimi_Sales/Code/outputs/core_simi_260501/logs/run_core_simi_story_exploration_260524.log
  log type:  text
 closed on:  26 May 2026, 14:46:40
------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------
```

这个和 review amount 挂钩，但差评越多就越多